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Additional Information

This issue of *SupBiotech Projects* contains research articles written by students enrolled in the Apprenticeship engineering program at SupBiotech, as part of their contribution to the Undergraduate Research Program in Bioengineering (5D Projects), supervised by Julien Grimaud, PhD. For more information, please contact Dr. Grimaud using the following address: julien.grimaud [at] supbiotech.fr

Editorial

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Welcome to Volume 8 of *SupBiotech Projects*! The articles featured in this volume mark the culmination of the 5D Projects, a hallmark the Apprenticeship program at SupBiotech. This predoctoral research program, rooted in the 5D methodology - Discover, Define, Design, Develop, Deploy - empowers our students to explore cutting-edge topics, develop solutions, and demonstrate their creativity, teamwork, and technical skills.

Each group has invested significant effort into all stages of their research journey, from conducting in-depth scientific reviews to exploring economic viability, gathering experimental data, and, ultimately, writing comprehensive scientific articles. As always, the papers in this journal have undergone a thorough peer-review process conducted by fellow SupBiotech students.

This year, we are excited to introduce a new tradition: the addition of visual abstracts to each published article. These visually engaging summaries provide a snapshot of the key concepts, methodologies, and outcomes of each project, making complex scientific findings more accessible to a broader audience.

We extend our heartfelt thanks to everyone who contributed to the creation of this journal. Special recognition goes to the Office of Apprenticeship Student Affairs, the Teaching Laboratories team, and the Research Laboratories staff for their support throughout the year. A special thanks to Claire Grassi for her invaluable input during the market research phase, laboratory manager Nedra Liamini for her dedication to ensuring the success of these projects, as well as laboratory tutors Jacqueline Bert and Nassim Arouche for overseeing the experimental part of the 5D Projects.

Lastly, we extend our deepest gratitude to the Apprenticeship class of 2025. Your dedication, creativity, and perseverance have not only brought your projects to life but have also made an enduring impact on the 5D Projects and the SupBiotech community. We look forward to seeing where your scientific journeys will take you.

Here's to many more breakthroughs in the future! May the discoveries within these pages inspire you and others to keep pushing the boundaries of science.

Evaluating the specificity and efficacy of purified V_HH antibody fragment targeting GFAP in astrocytes and brain organoids

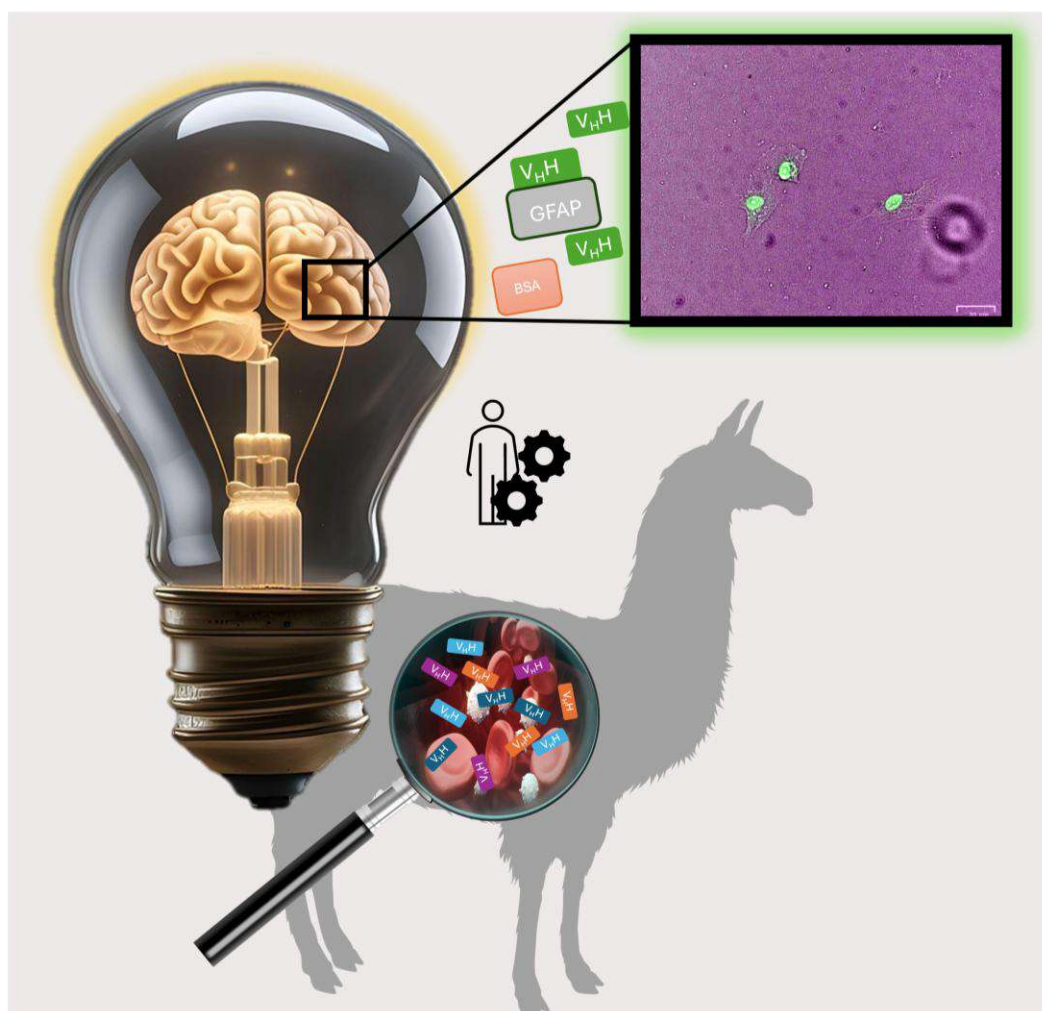
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VISUAL ABSTRACT



Evaluating the specificity and efficacy of purified V_HH antibody fragment targeting GFAP in astrocytes and brain organoids

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Abstract – The Glial Fibrillary Acidic Protein (GFAP) plays a crucial role in maintaining astrocytic integrity and serves as a biomarker for neuroinflammation and neurodegenerative diseases such as Alzheimer's. However, conventional antibodies face challenges in targeting GFAP within the central nervous system due to their inability to cross the blood-brain barrier. This study evaluates camelid-derived antibody fragments, known as Variable Heavy-Chain domains (VHH), tagged with polyhistidine (polyHis), for their specificity and binding efficiency to GFAP. Using techniques such as SDS-PAGE, ELISA, and immunofluorescence, these nanobodies demonstrated strong tissue penetration, high specificity, and effective binding to GFAP in astrocytic cells and brain organoids. Unexpected observations, such as the nuclear localization of GFAP under inflammatory conditions, highlight the potential existence of alternative forms of GFAP or interactions with other proteins, suggesting new directions for future research. These findings confirm the diagnostic potential of VHH fragments and pave the way for their application in the early detection and imaging of neurodegenerative processes. Enhancing these nanobodies with radiotracers could further expand their utility for real-time diagnostic monitoring, offering a promising tool to advance the diagnosis and treatment of central nervous system pathologies.

Index Terms – Astrocytic cells, Brain organoids, Glial Fibrillary Acidic Protein (GFAP), Nanobodies, Neurodegenerative diseases, Variable Heavy-Chain domains (VHH).

INTRODUCTION

The Glial Fibrillary Acidic Protein (GFAP) is an intermediate filament protein, located in the cytoskeleton. This filamentous protein is found in glial cells and mainly in astrocytes, playing

a crucial role in maintaining cytoskeletal integrity and mechanical stability^{1,2}.

While primarily associated with the central nervous system, GFAP expression extends to other glial populations, including enteric glial cells and Schwann cells within the peripheral nervous system¹.

The GFAP gene, located on human chromosome 17q21, encodes several isoforms, the most common of which is GFAP- α . Post-translational modifications, influenced by kinases, growth factors and hormones, regulate its expression¹.

Dysregulation of GFAP expressions or mutations in the gene are associated to several neurodegenerative diseases. Elevated levels of this protein often serve as a biomarker, notably in conditions such as Alzheimer's, Alexander disease or traumatic brain injury^{1,3}. Consequently, GFAP serves as a biomarker for astrocyte activation and neuroinflammation, enabling the early detection of astroglial alterations preceding clinical symptoms¹. Under physiological conditions, GFAP is primarily localized within the nucleus, whereas during inflammatory states, it shifts to a perinuclear localization, reflecting changes in astrocyte reactivity⁴.

Currently, the diagnosis of neurodegenerative diseases relies primarily on the manifestation of clinical symptoms, which are often detected at an advanced stage when neuronal damage is already significant. However, molecular and biochemical alterations in the central nervous system preceded these symptoms, underscoring the need for precise molecular tools for early diagnosis. Various studies are underway, exploring the plasmatic detection of neuroinflammation or biomarkers of disease precursors^{5,6}. Conventional antibodies, usually used for labeling, can't effectively target GFAP within the brain due to their large

size, which prevents them from crossing the blood-brain barrier.

Camelid-derived antibody fragments, known as Variable Heavy-Chain domains (V_HH), present a promising approach for GFAP detection. Unlike conventional antibodies, the V_HH are devoid of light chains and possess two heavy chains, having a small size (12-15kDa)⁷. These characteristics give them high stability and affinity for their target antigen, especially thanks to extended CDR3 residues⁸. Thanks to their small size they can penetrate tissues normally inaccessible to conventional antibodies, such as the blood-brain barrier⁹.

These properties enable V_HH fragments to bind to complex or less accessible structures, making them particularly suited for applications in the diagnosis and treatment of central nervous system (CNS) diseases, where they can overcome barriers such as the blood-brain barrier. Additionally, the incorporation of a polyhistidine (polyHis) tag enhances the purification, detection, solubility, and immobilization of these fragments¹⁰. Recent advancements in nanobody research confirm this approach. Pierre Lafaye and his team at the Pasteur Institute have made significant contributions to the development of nanobodies, specifically on antibody engineering and the production of V_HH s targeting specific proteins like GFAP¹⁰. Their work includes generating V_HHs from camelids to target GFAP, with potential applications in diagnostics, such as magnetic resonance imaging -based imaging for Alzheimer's disease¹¹. Additionally, they are exploring the therapeutic use of V_HHs, for example, in treating hemophilia¹² and diagnosing zoonotic diseases¹³.

The following work aims to evaluate the specificity and binding efficiency of polyhistidine-tagged V_HH fragments specifically targeting GFAP. Building on the preliminary work conducted by Dr. Lafaye and collaborators, these V_HH antibodies have been tested on astrocytes and brain organoids. Qualitative methods including SDS-PAGE, ELISA, and immunofluorescence were employed to confirm the specificity of the polyHis-labeled V_HH binding, providing a comprehensive assessment of their diagnostic potential.

MATERIAL AND METHOD

CONSTRUCTION OF A V_HH LIBRARY

The plasmid for V_HH polyHis-tag expression was constructed as described in the article¹⁰. The V_HH coding sequence was subcloned into the pET22 vector using NcoI and NotI restriction sites. This allowed expression of the V_HH protein with a polyHis-tag at the carboxy-terminal end.

The full sequence of the VHH polyHis-tagged in pET22 plasmid is:

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CCATGGCTCAGGTTACAGTCCAGGTGAAAGGTGGTGGCAAAGTGCAACCTGGCGGTAAGCT
GCGTTTGCGCTGTGACAGCTTCTGGCTTACGTTCTCGAACTATGCGATGACTGGTTACGCCA
AGCACCAGGGAAAGGCATCGAATGGGTAAGCCGCAATTGGACCGGTGGAAAGCCATACGGA
GTATGCGGATTCCGTTGAAAGGCCGCTTTCGCAATTCGTCGTGACAAATGCGAAGAACACCTGA
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GTCTCGGGATGCGCAAACCTGAAACCGGAAGATACCGCGCTCTACTATTGCGCCATCACTCAG
CAGGGGTACAGGTCGTGGCCAAAGGCACACGTGTACCGTTTCCAAAGCGGCCGCACTCGAGC
ACCACCACCACCACTGAGATCCGGTGCTACAAAGCCGGAAGGAAGCTGAGTTGGCT
GCTGCCACCGCTGAGCAATAACTAGCATAACCCCTTGGGGCTCTAAACGGGTCTTGAGGGG
TTTTTGTGAAAGGAGGAACTATATCCGATTGGCGAATGGGACGCGCCCTGTAGCGGCG
CATTAAAGCGCGGGGTGGTGGTTACGCGCAGCGTGACCGCTACACTTGGCAGCGCCCT
AGCGCCCGCTCTTTGCTTCTTCCCTTCTTCTGCGCACGTTGCGCGGCTTCCCGCTCAAG
CTCTAAATCGGGGCTCCCTTATAGGTTCCGATTAGTGCTTACGCGACCTGCAGCCCAAAA
AACTTGATTAGGGTGATGGTTACGTAGTGCGCCATCGCCCTGATAGACGGTTTTTCGCCCTT
TGACGTGGAGTCCACGTTCTTAAATAGTGGAAGTCTTGTCCAACTGGAACAACACTCAACC
CTATCTCGGTCTATTCTTTGATTATAAGGGATTTCGCGATTCGCGCTATTGGTTAAAAAT
GAGCTGATTAAACAAAATTTAACGCGAATTTAAACAAAATATTAACGCTTACAATTTAGGTG
GCATTTTTCGGGAAATGTGCGCGGAACCCCTATTGTATTCTTCTAAATACATTCAAATAT
GTATCCGCTCATGAGACAATAACCTGATAAATGCTTCATAAT
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Expression and purification were carried out according to established protocols. The plasmids containing the V_HH sequences were verified by sequencing using T7 promoter and terminator primers.

BL21 STRAINS

The expression of anti-GFAP VHH was made possible by *Escherichia coli* BL21 (DE3) pLysS cells. These were supplied already transformed by heat shock by Dr. Pierre Lafaye's team at the Pasteur Institute¹⁴. The cells were cultivated in liquid Luria-Bertani (LB) broth at a concentration of 20 g/L and ampicillin was added at a final concentration of 100 µg/mL to maintain selective pressure for plasmid retention.

TISSUES - HUMAN ORGANOID AND HUMAN ASTROCYTES CULTURES

The astrocyte and human organoid samples were supplied by the CellTechs laboratory at the SupBiotech engineering school^{15,16}. The organoids were obtained in vitro by neuronal differentiation of healthy pluripotent stem cells and the astrocytes supplied were also healthy. Sections of 20 µm organoids were cut in a cryostat in the presence of OCT (cryogenic embedding medium). Human astrocyte cells were cultured in well slides. Both were fixed with 4% paraformaldehyde (PFA) prior to labelling.

EXPRESSION OF ANTI-GFAP V_HH

The expression of anti-GFAP V_HH antibody fragments by BL21 bacteria was induced by adding isopropyl β-D-1-thiogalactopyranoside (IPTG) to the culture medium, following the protocol previously described by Dr. Pierre Lafaye¹⁴. Transformed BL21 colonies were inoculated into LB liquid medium (Sigma, L3022-1KG) with ampicillin (Sigma, A9518-25G) overnight, at 37°C under agitation. The preculture was then inoculated into LB Ampicillin liquid medium (1/100 dilution) and kept under agitation 150 rpm at 37°C until reaching an OD of 0.5. BL21 growth monitoring was measured with a spectrophotometer (Jenway, 7315) at 600 nm. When the desired cell density was reached, IPTG was added to a final concentration of 0.5 mM. The culture was incubated overnight at 30°C, shaking at 150 rpm. IPTG will act as an inducer of VHHs expression.

EXPRESSION OF ANTI-GFAP V_HH

For periplasmic extraction, bacteria were centrifuged at 8000 rpm for 10 min. After cultures centrifugation, the cell pellet was resuspended in resuspension solution (contains 0.5 M of NaCl and 0.29 M of Na₂HPO₄ 2H₂O). The Polymyxin B solution (Sigma Aldrich, Fluka 81334) was added to reach a concentration of 1 mg/mL. The mixture was incubated for 1 hour at 4°C, then centrifuged for 10 min at 80000 rpm and the supernatant was collected.

PURIFICATION OF ANTI-GFAP WHH WITH NINTA BEADS

Purification of polyHis-labelled nanobodies was performed by affinity chromatography on a Ni-NTA resin. The resin was first prepared by homogenising the Ni-NTA agarose suspension, followed by centrifugation at 500 g for 5 min to remove the storage solution. The resin was then equilibrated with 10 volumes of binding buffer (50 mM NaH₂PO₄, 300 mM NaCl, 10 mM imidazole, pH 8.0), this step was repeated twice. The protein extract containing the nanobodies was incubated with the resin under gentle agitation (200 rpm) for 1 hour at room temperature. After centrifugation (500 g, 5 min), the supernatant was collected for analysis. The resin was washed three times with wash buffer (50 mM NaH₂PO₄, 300 mM NaCl, 20 mM imidazole, pH 8.0) prior to stepwise elution of nanobodies with increasing concentrations of imidazole (20, 50, 100, 250 and 500 mM) in 300 mM PBS-NaCl buffer pH 8.0. Each fraction was collected separately and stored at 4°C for further analysis by SDS-PAGE, ELISA and imagery.

SDS-PAGE CONTROL

To verify that the purification of our anti-GFAP V_HH fragment was optimal, an SDS-PAGE gel was run. Harvested samples were prepared by adding 1/6 dye 6X loading gel (New England Biolabs, B7024S), then heated at 95°C for 5 min. They were then loaded in a triplicate 10% SDS-PAGE gels. Samples migrate in 1X TGS buffer (Sigma, T5912-1L) for 30 min at 80 V, then at 110 V. The gels were rinsed with distilled water and stained with Coomassie blue for 15-20 min under gentle agitation. The gels were then washed with destaining buffer (10% ethanol, 10% acetic acid).

ELISA FOR V_HH SPECIFICITY CONTROL

The objective is to demonstrate the specificity of the anti-GFAP V_HH polyHis-tag. The plate was coated with GFAP 1 µg/mL (preparation previously described¹⁰) and BSA 1 mg/mL (Sigma-Aldrich, 810533) in 1X PBS and incubated overnight at 4°C. The following day, excess liquid was removed, and the wells were washed with washing buffer (0.05% Tween20, Sigma-Aldrich P1379). V_HH samples (different dilutions) and calibration range were added in duplicate, and the plate was incubated for 1 hour at 37°C. Between each step, the wells were washed with the washing

buffer. Next, to detect V_HH, anti-HIS-peroxidase + HRP secondary antibody (1 µg/mL) was added and incubated for 1 hour at 37°C with gentle agitation. OPD substrate (1 mg/mL) was then added and incubated for 40 min. The reaction was stopped by adding 3M hydrochloric acid solution (Sigma-Aldrich, 258148). Relative fluorescence units were measured by spectrophotometry (Multiskan Sky High, Thermofischer Scientific) at 492 nm after excitation at 355 nm. The concentration of each well was determined using the equation for the calibration line. For the ELISA method, duplicates were performed for each sample to assess measurement consistency and limit technical variations, ensuring better experimental reproducibility.

V_HH FUNCTIONALITY TEST

To test the functionality, anti-GFAP polyHis-tag V_HHs were tested on paraffin-embedded brain sections and/or organoids. Non-specific binding sites were blocked with a solution of 2% bovine serum albumin (Sigma, 810533), 0.5% Tween 20 (Sigma, P1379-100ML) for 2 hours. The wash solution used between all subsequent steps was TBS, 0.5% Tween 20. For astrocytes, the anti-GFAP V_HH primary antibody produced (1 µg/mL) was incubated overnight in a humidified chamber at room temperature. Fluorescent anti-His secondary antibodies were incubated for 2 hours at room temperature before being observed under a ZOE microscope (ZOE Fluorescent Cell Imager, BIORAD).

For organoid sections, the primary anti-GFAP V_HH antibody produced (1 µg/mL) and for control sections, the anti-GFAP polyHis-tag antibody was incubated overnight in a humidified chamber at room temperature. Secondary anti-His antibodies conjugated to rabbit streptavidin/HRP (dilution 1/3000 in the wash solution) were applied to the sections and incubated for 2 hours at room temperature. The sections were then placed in a 2% Harris haematoxylin bath (Labkem, HEMA-HPS-500) for 5 min. Organoid sections were observed under an inverted light microscope between slides and coverslips mounted with Mowiol.

RESULTS*PURIFICATION OF PERIPLASMATIC EXTRACTS AND CONTROL OF V_HH FRAGMENTS PURITY.*

The purification of periplasmic extracts by immobilized metal affinity chromatography (IMAC) resulted in a fraction containing the V_HH fragments, specifically the fraction with 100 mM imidazole. The purity and homogeneity of the V_HH were confirmed by SDS-PAGE.

Controls were performed using the initial eluate containing PBS-NaCl (300mM) and Imidazole (50mM) (Figure 1.a), as well as a negative control with loaded water (Figure 1.a). These tests were conducted to determine whether the buffer solution had any impact on the experiment or contains any contaminants. The loaded water control ensured that any

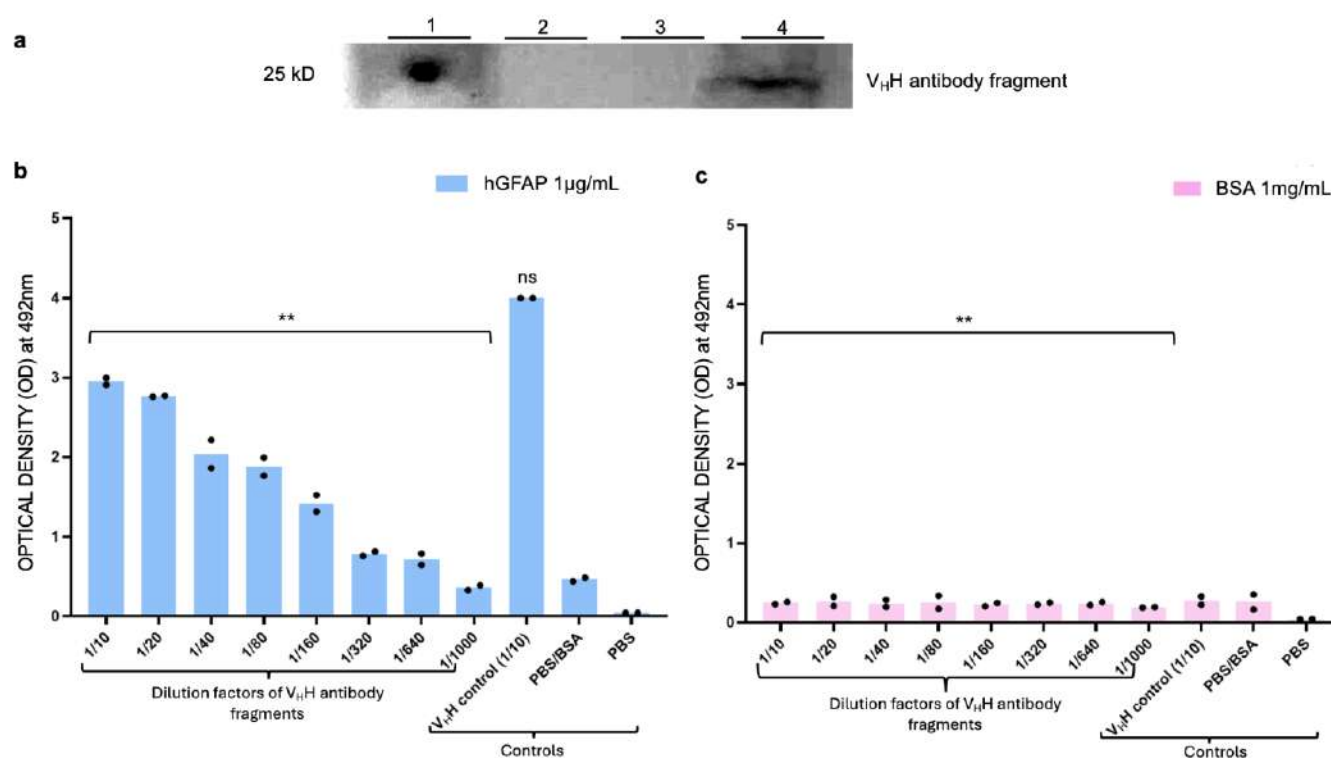


Figure 1: Analysis of VHH antibody fragments targeting GFAP: SDS-PAGE and ELISA characterization. (a) SDS-PAGE analysis of VHH antibody fragment. A polyacrylamide gel electrophoresis (SDS-PAGE) was performed to assess the purity of VHH antibody fragment targeting GFAP, followed by Coomassie blue staining for protein visualization. Each sample was loaded in triplicate to ensure reproducibility. The individual wells were loaded as follows: (1) Molecular weight marker (Precision Plus protein dual color standards: 10-250 kDa); 20µL. (2) Negative control: 17µL of loaded water + 3µL of gel loading dye (loaded in triplicate). (3) Initial eluate: 17µL of the fraction collected after elution with PBS-NaCl (300mM) and Imidazole (50mM), mixed with 3µL of gel loading dye (loaded in triplicate). (4) Purified protein fraction: 17µL of the proteins sample obtained after Ni-NTA affinity purification using 100mM Imidazole, mixed with 3µL of gel loading dye (loaded in triplicate). (b) Qualitative Indirect ELISA Measurements of Human GFAP Specificity for VHH Antibody Fragments. This graph represents the optical density (OD) values measured for VHH antibody fragments when exposed to human GFAP (hGFAP) at different dilution factors. The mean OD values for each dilution are displayed in the histograms. A Wilcoxon unpaired-data test was performed on $n = 16$ samples, showing a significant difference (p -value = 0.007813, **). While a Wilcoxon paired-data test was conducted on $n = 2$ samples and revealed no significant difference (p -value = 1, ns). (c) Qualitative Indirect ELISA Measurements of Human GFAP Specificity for VHH Antibody Fragments. In contrast to graph (b), it presents the OD values when VHH antibody fragments were exposed to Bovine Serum Albumin (BSA) instead of hGFAP, also measured at different dilution factors. The histograms display the mean OD values for each dilution, with the Wilcoxon unpaired-data test performed on $n = 16$ samples showing a significant difference (p -value = 0.007813, **). Controls performed in the ELISA assay: Positive control: VHH antibody fragments + hGFAP; Negative controls: VHH antibody fragments + BSA; PBS/BSA + hGFAP or PBS/BSA + BSA; PBS + hGFAP or PBS + BSA.

observed signal or contamination was not caused by the reagents or the equipment itself. These controls allowed the verification of the results' reliability ensuring that the observed signal was caused by the presence of V_HH fragments and not by the buffer solution or any external contamination.

As expected, no band was observed in the controls. In contrast, a band corresponding to V_HH fragments was detected at 25 kDa.

TESTING THE SPECIFICITY OF VHH FRAGMENTS FOR hGFAP.

The specificity of the nanobodies for GFAP was tested using a qualitative indirect ELISA measurement of human GFAP (Figure 1.b, Figure 1.c). Optical density (OD) was measured

at 492nm because it corresponds to the absorption wavelength of the colored product generated by the oxidation of OPD. Values of OD decreased progressively in correlation with the dilution factor of V_HH antibody fragments produced (and so the concentration of V_HH antibody fragments) in hGFAP-coated wells, following a sigmoidal curve (Figure 1.b). Indeed, at low concentrations of V_HH antibody fragments, only a small quantity of antibodies is fixed, resulting in a low OD. At intermediate concentrations, a large number of antibodies are captured, causing the OD to increase rapidly. At high concentrations, the saturation of binding sites limits the enzymatic reaction, leading the OD to reach a plateau (Figure 1.b). The corrected mean OD was calculated as the average OD of duplicate test measurements minus the average

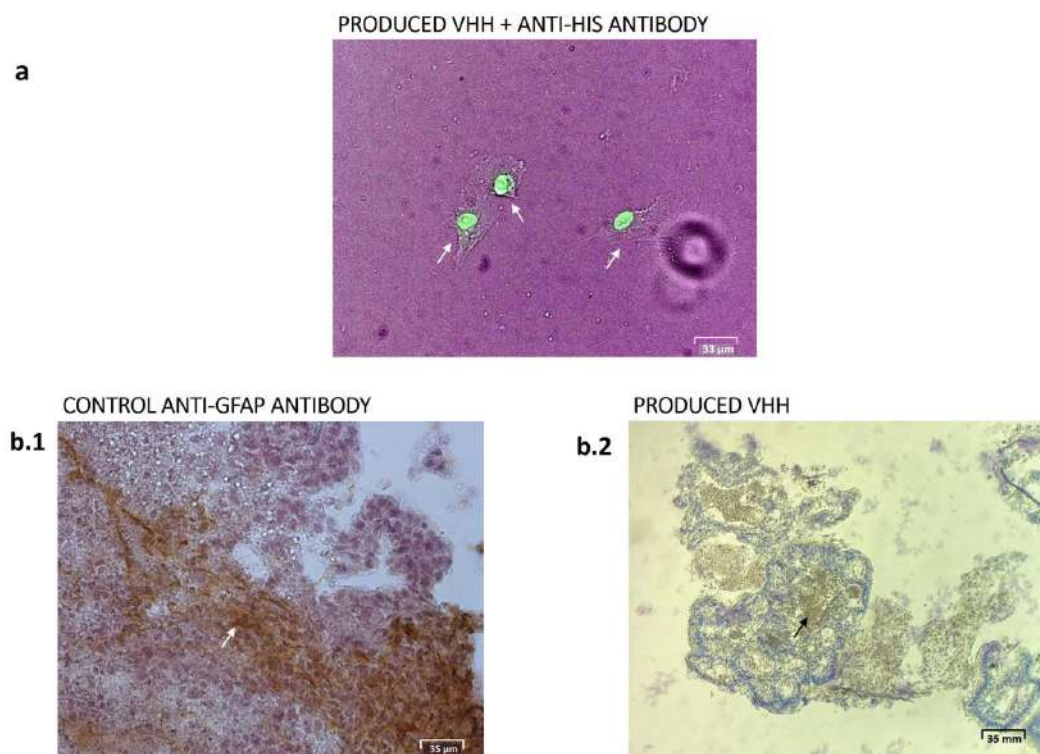


Figure 2: Validation of the effectiveness of the produced nanobodies targeting GFAP. (a) Fluorescence microscopy image of astrocytes labeled with the created VHH fragment, specifically targeting GFAP protein coupled to a His-tag, with detection by a GFP-conjugated anti-His secondary antibody for fluorescence visualization. The arrow indicates the labeled astrocytes, while the green corresponds to the GFAP targeted by the VHH produced, here in perinuclear and cytoplasmic location. (b) Optical images of immunohistochemically labelled brain organoid sections. In image 2.b.1, detection of GFAP with an anti-GFAP antibody used as a control; in image 2.b.2, detection of GFAP with the VHH fragment produced. In both images, the brown staining indicates GFAP labelled with the elements described above with a reaction between the streptavidin-coupled anti-His secondary antibody and the HRP enzyme. The purple staining represents the remaining tissue and cell structures revealed by the Harris haematoxylin counterstain. The arrows show part of the brown staining of the GFAP in question.

OD of duplicate PBS controls (Figure 1.b, Figure 1.c). This calculation allows for the elimination of external interferences such as substrate absorption or instrument effects. The corrected mean OD for V_HH antibody fragments produced diluted at 1/10 was 2.908, whereas for those diluted at 1/640, the corrected mean OD was 0.672 (Figure 1.b). For OD values of V_HH dilutions in BSA-coated wells, no significant values were observed, with no correlation (Figure 1.c). In other words, this observation means that their corrected mean OD values are below the threshold (0.419, defines by the test hGFAP + PBS/BSA), as well as the negative controls (corrected mean OD of V_HH antibody fragments controls + BSA = 0.233, corrected mean of PBS/BSA + BSA = 0.216).

An experiment testing the sensibility of V_HH antibody fragments for BSA was conducted under the same conditions as the experiment testing the sensibility of V_HH antibody fragments for hGFAP (Figure 1.c). This test (V_HH antibody fragments + BSA) was performed to confirm the specificity of V_HH antibody fragments for hGFAP and to show the absence of non-specific interactions with another protein as well as the absence of experimental artefacts.

The corrected mean OD values for our control V_HH on hGFAP and BSA, we obtain very different measurements, respectively: 3.956 and 0.233 (Figure 1.b, Figure 1.c).

A Wilcoxon test on unpaired data revealed that the corrected mean ODs corresponding to dilutions of V_HH fragments in hGFAP-coated wells were significantly different (p-value = 0.000931, **) from those in BSA-coated wells (Figure 1.b, Figure 1.c). This test used unpaired data because it compares two independent groups: detection of hGFAP versus detection of BSA. The results confirm that V_HH fragments specifically bind to hGFAP rather than BSA.

A second Wilcoxon paired-data test revealed no significant difference (p-value = 1, ns) between the corrected mean OD of the V_HH antibody fragments produced for detecting hGFAP (diluted at 1/10) and that of the control test, where control V_HH antibody fragments (diluted at 1/10) (provided by P. Lafaye, Institut Pasteur) were exposed to hGFAP (V_HH controls + hGFAP) (Figure 1.b). This statistical test used paired data to compare the affinity of the experimentally produced V_HH antibody fragments with that of the control V_HH antibody fragments for hGFAP. The results indicate that

the produced V_HH antibody fragments exhibit a similar affinity for hGFAP as the control V_HH antibody fragments.

IMMUNOHISTOCHEMISTRY AND IMMUNOFLUORESCENCE

Analysis of immunohistochemical and immunofluorescent staining reveals different types of staining. Image 2.a, illustrating the results of immunofluorescence staining of astrocytes, shows three cells of characteristic “star” morphology. A marked concentration of green fluorescence is observed around the nuclei of these glial cells, accompanied by very slight, barely perceptible diffusion in the rest of the cell.

Both said images 2.b, presenting immunohistochemical labeling of organoid sections, reveal two distinct observations, influenced by brightness, contrast, as well as the magnification used. The higher magnification provides more cellular detail on the control sample than on the V_HH -treated sample. However, the staining profile remains similar in both cases. This is characterised by brown areas, indicating labelling of GFAP using an anti-GFAP antibody for the control and V_HH produced for the test. These markers are fixed by an anti-his antibody coupled to streptavidin, allowing a reaction with HRP. The purple areas correspond to the rest of the tissue, the cells and molecules, highlighted by counterstaining with Harris haematoxylin. On both images, control (Fig. 2.b.1) and V_HH (Fig. 2.b.2), brown areas are distributed within the purple staining, sometimes between the purple structures and sometimes directly on them. These images show a marked difference in the intensity and distribution of the brown marking between the control sample and that treated with V_HH.

DISCUSSION

Our study provides key information on the production, purification, and functionality of V_HH antibody fragments targeting hGFAP and carrying a polyHis-tag. Our results complement the work of Perruchini et al., 2009¹⁰, which demonstrated the advantages of V_HH antibody fragments. Their superior tissue penetration compared to conventional monoclonal antibodies is the main point highlighted in this article.

PURIFICATION AND VERIFICATION OF THE PURITY OF V_HH ANTIBODY FRAGMENTS

To begin, the purification of periplasmic extracts by affinity chromatography on immobilized metal (NiNTA beads) allowed the isolation of the V_HH antibody fragments. The observation of a well-defined band at 25 kDa by SDS-PAGE confirmed the presence of the V_HH antibody fragments in the eluate of interest, by its characteristic size. The absence of bands in the negative controls, charged water and the initial eluate validated the purity of the V_HH antibody fragments and the efficiency of our production technique. The subsequent specificity and functional tests, such as ELISA and

immunolabelling, require a high degree of purity to ensure that the interactions are specific and therefore solely attributable to the V_HH fragments.

V_HH ANTIBODY FRAGMENTS' SPECIFICITY AND SENSIBILITY

Following purification, the high specificity and sensitivity of the V_HH antibody fragments for hGFAP were confirmed using the qualitative indirect ELISA technique. The progressive decrease in the corrected mean OD, observed in correlation with the dilution factor of the V_HH antibody fragments, suggests that dilution impacts the detection of hGFAP by these fragments. This result seems logical and lends credibility to our subsequent ELISA results.

Furthermore, when we compare dilutions of V_HH produced on hGFAP and BSA, we see a very different trend, with very low OD results, even negligible, for V_HH produced and BSA. This shows that our antibody has no affinity for BSA, regardless of the concentration of V_HH produced. This is confirmed by our Wilcoxon test on unpaired data with a p-value of 0.000931, showing a clear difference between the two data. This demonstrates that our V_HH product is specific to hGFAP and not to BSA. Better specificity can be demonstrated by ELISA testing of our V_HH on GFAP-like molecules, such as vimentin¹⁷.

The corrected mean OD for the control (PBS/BSA + hGFAP) corresponds to the value obtained when no V_HH antibody fragments are in contact with hGFAP, which is 0.419. This value serves as the baseline for comparison. The 1/640 dilution of V_HH antibody fragments, with a corrected mean OD of 0.672, is identified as the threshold dilution, as it exceeds the baseline. In contrast, the 1/1000 dilution exhibits a corrected mean OD of 0.315, which is below the baseline corrected mean OD defined by the control (PBS/BSA + hGFAP).

Therefore, the 1/640 dilution represents the maximum dilution of the V_HH antibody fragments that still allows for the specific detection of hGFAP. Initially, the initial concentration was $3.2 \cdot 10^{-2}$ mg/mL so the minimum functional concentration of approximately $5 \cdot 10^{-5}$ mg/mL demonstrates an extremely low sensitivity level, which is sufficient for the specific detection of human GFAP.

After observation of the V_HH controls on hGFAP and BSA, a marked difference between the optical densities (OD) was observed, indicating that the V_HH directed against GFAP has no affinity for BSA. This confirms that the V_HH produced does not interact with BSA.

Statistical analysis (Wilcoxon paired test) shows that the corrected mean OD associated with V_HH antibody fragments diluted at 1/10 was not significantly different from those observed under control conditions (V_HH antibody fragments controls + hGFAP). These tests reveal no significant difference between our observations, suggesting that our

produced V_HH shows a similar level of reaction to that of the control V_HH.

However, as the experiment was conducted with a single duplicate per condition tested, we cannot assert this conclusion. To validate these results, it is crucial to repeat the experiment with a larger number of replicates, which would enable a robust statistical analysis.

Surface Plasmon Resonance (SPR) would enable real-time measurement of the affinity and interaction kinetics of V_HH antibody fragments with GFAP, facilitating estimation of their binding strength and determination of their affinity constant¹². A cross-reactivity assay could also be performed to assess V_HH reactions in the presence of GFAP-bound proteins or human proteins, allowing non-specific interactions to be ruled out.

IMMUNOLABELLING: CONFIRMATION OF CELLULAR AND TISSUE TARGETING BY VHH ANTIBODY FRAGMENTS

The immunofluorescence of astrocytes (Figure 2.a) highlights several features of our experiment. First and foremost, the presence of fluorescence confirms that our experiment, under the conditions defined above, was successful. Secondly, the specific localization of this fluorescence provides important information on the characteristics of our V_HH. It demonstrates a unique affinity for cellular elements, with no significant interaction with other environmental components or elements associated with astrocyte attachment.

Furthermore, the distribution of fluorescent labeling, characterized by increased intensity around the nucleus and lower diffusion into the cytoplasm, strongly suggests that our V_HH targets the GFAP of human astrocytes. This perinuclear localization is consistent with the functional and structural role of GFAP in this region, where it forms a network of intermediate filaments around the nucleus. This network contributes to the mechanical stability of cells and the organization of intracellular organelles. Furthermore, the observed cytoplasmic diffusion reflects GFAP's involvement in the cytoskeleton for the overall structure of the astrocyte, playing a key role in maintaining its architecture and interactions with the cellular environment^{18,19}.

Immunohistochemical labelling (figure 2.b) of the two images obtained allows several relevant observations to be made. The differences between the images can be attributed to variations in brightness, contrast and magnification, as well as different organoid sections and experimental manipulations. However, despite these variations, a similar pattern is observed in both cases: cells and other tissue structures appear purple-stained, using Harris haematoxylin, with areas of varying darkness. Whereas areas of brown staining, revealing GFAP, following the reaction of the anti-His secondary antibody coupled to streptavidin and the enzyme HRP. These brown areas are visible between or on top of purple areas or dark purple structures. These results confirm that our V_HH acts in a

similar way to the anti-GFAP control antibody, targeting the same structures in comparable samples and thus confirming its affinity for GFAP. In addition, the images confirm the observations obtained by immunofluorescence, revealing a characteristic concentration of brown staining around darker purple circles, presumably the nuclei. This distribution suggests an accumulation of GFAP in the perinuclear zones, as well as in the astrocytic extensions surrounding the cells.

These analyses underline the specific affinity of our V_HH for GFAP in human astrocytes. The results show a characteristic perinuclear localization, consistent with the structural and functional role of GFAP, as well as a distribution in astrocytic extensions, contributing to cell architecture. Once V_HH has been confirmed to function correctly by co-localization experiments with a GFAP-binding control, it would be useful to evaluate its ability to cross the blood-brain barrier.

CONCLUSION

This study demonstrates the effectiveness of V_HH antibody fragments targeting GFAP for the specific detection of GFAP protein in astrocytic cells and brain organoids. Thanks to their small size and stability, these nanobodies exhibit high specificity for GFAP, as confirmed by techniques such as ELISA and microscopy imaging. The penetration through the blood brain barrier and tissues remains to be confirmed. These findings validate the production and purification protocol of V_HH fragments, as well as their functionality in an experimental context. After confirming the proper functionality of the V_HH, it would be important to investigate their ability to cross the blood-brain barrier. This could be done using reconstituted blood-brain barrier models *in vitro* or through *in vivo* studies in mice, followed by post-mortem analysis to assess their penetration and distribution within the brain tissue.

The results show that GFAP is mainly located around the nucleus, which aligns with its role in maintaining the structural integrity of the cell. Additionally, GFAP is found in astrocytic extensions, where it helps shape the overall cell structure. These findings emphasize the key role of GFAP in organizing the astrocyte, especially in forming the network of intermediate filaments around the nucleus and supporting the cytoskeleton to maintain the cell's architecture.

Finally, this innovative approach, based on the use of V_HH antibody fragments, could represent an important step toward the development of early diagnostic tools for neurodegenerative diseases, such as Alzheimer's disease, where early detection is critical because of the complexity to penetrate deep brain tissues, and the lack of potential biomarker. To further enhance their clinical utility, future studies should investigate the potential of labeling these nanobodies with radiotracers, enabling their application in brain imaging for diagnostic monitoring and real-time evaluation of neurodegenerative processes. The potential applications of these nanobodies in diagnosing and treating

central nervous system pathologies deserve further exploration.

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AUTHORS' CONTRIBUTIONS

L.C. and M.P. wrote the Materials and Methods section of the article. C.G. and C.Rou. wrote the Results and Discussion section of the article. E.N. and C.Rop. wrote the Introduction section of the article. J.V. wrote the Conclusion section of the article. J.P. wrote the Abstract section of the article.

CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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Improved bioproduction of prodigiosin by *Serratia marcescens* using peanut powder via a low-cost process

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Design of a mycelium-based insulating biomaterial: a sustainable and more efficient insulation

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Combating resistant bacteria: exploring the synergy of essential oils and antibiotics

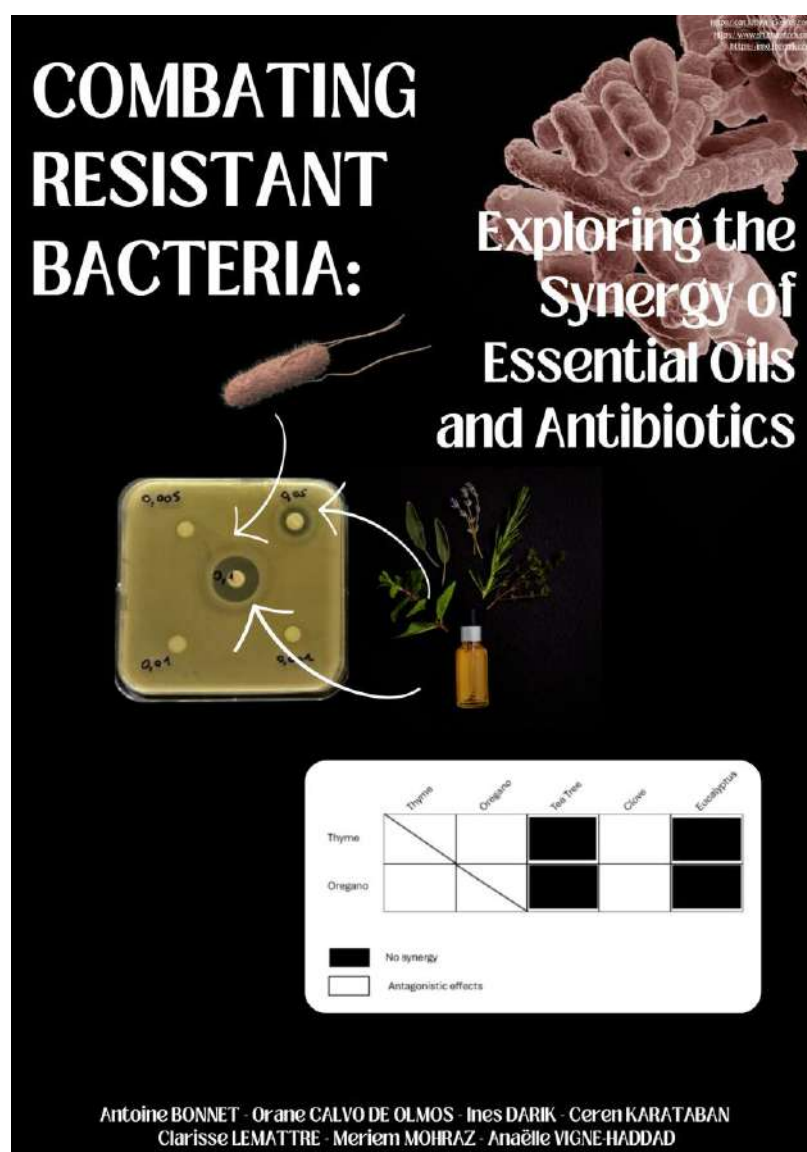
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VISUAL ABSTRACT



Combating resistant bacteria: exploring the synergy of essential oils and antibiotics

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Abstract – Antibiotic resistance represents a growing threat to global public health, largely due to the rapid spread of multi-resistant bacteria such as those in the Enterobacteriaceae family. In this study, we investigated the antimicrobial potential of certain essential oils (EOs)-including tea tree, clove, thymol, oregano and eucalyptus-used alone or in combination with antibiotics, using *E. coli* as a model.

We used disc diffusion assays and Minimum Inhibitory Concentration (MIC) measurements to demonstrate that these EOs can significantly inhibit bacterial growth, sometimes comparable to conventional antibiotics. We tried to identify synergistic interactions between EOs and antibiotics. Our results also show that combining EOs with antibiotics could reduce the doses required, thus limiting the selective pressure that leads to the emergence of resistance. In addition to their antibacterial properties, these essential oils have the advantage of being less expensive and more accessible, making them promising candidates as alternatives or supplements in the fight against bacterial resistance.

Index Terms – Antibiotic, Antibiotic resistance, *Escherichia Coli*, Essential oil, Synergy

INTRODUCTION

The treatment of bacterial infections with antibiotics is a major public health issue. First introduced in the 1930s, these molecules capable of inhibiting bacterial growth have saved millions of lives since the 1940s. However, their massive use and misuse have led to an increase in antibiotic resistance. This phenomenon affects underprivileged and developed countries alike, making it a global problem. To tackle this problem, the World Health Organization (WHO) published a list of antibiotic-resistant “priority pathogens” in 2017, to coordinate research and development of new antibiotics. In

this release, 12 strains are prioritized, with Enterobacteriaceae classified as a critical category priority¹.

Among these, *Escherichia coli* is of particular concern because of its role in nosocomial infections, but also urinary tract infections, pneumonia and septicemia².

Antibiotics are today's standard antibacterial treatments. They are the most widely used thanks to their multiple modes of action, enabling them to target a wide range of bacteria. Derived from micro-organisms such as fungi and bacteria, they have a bactericidal or bacteriostatic action. Antibiotic treatment has revolutionized the world of medicine, considerably increasing life expectancy³.

Some antibiotics inhibit cell wall synthesis, cell membrane function, or even protein or nucleic acid synthesis. The modes of action are multiple, with a common strategy: to impact the survival and multiplication of infectious bacteria⁴.

However, this treatment has one limitation: antibiotic resistance⁵. Antibiotic resistance is a major challenge for public health worldwide, particularly in the case of bacteria such as *Escherichia coli*. This phenomenon can be caused by a variety of mechanisms. The acquisition of resistance genes through horizontal gene transfer, often mediated by mobile genetic elements such as plasmids and transposons, remains one of the most widespread mechanisms. This process enables bacteria to acquire new resistance genes, whether or not they belong to the same species, via conjugation, transformation or transduction mechanisms⁶.

The most frequently observed resistance mechanisms in *E. coli* include the acquisition of extended-spectrum β -lactamases (ESBLs) and carbapenemases, responsible for resistance to β -lactam antibiotics, essential in clinical practice. In addition, more specific mechanisms such as efflux pumps and porin mutations enable *E. coli* to resist a range of

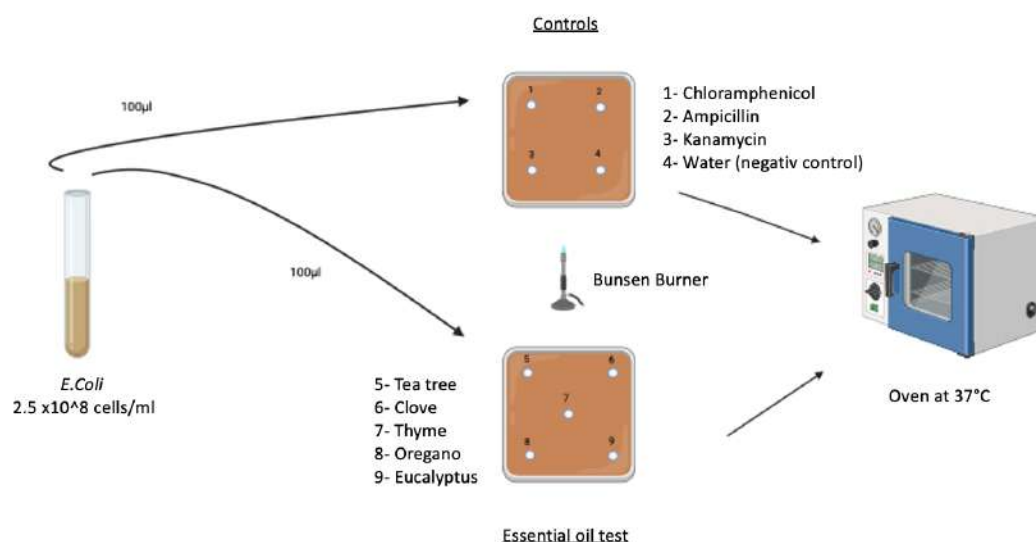


Figure 1: Schematic representation of the disc disposition plan.

antibiotics, including fluoroquinolones and aminoglycosides⁶.

Resistance can also result from the selective pressure exerted by the overuse of antibiotics, favoring the selection of resistant strains⁷. According to the “Santé Publique France” website, the number of people affected by antibiotic resistance in France in 2015 was close to 200,000, including 5,500 deaths⁸. These situations, which have an impact on the medical world, suggest the use of new treatments to address the problems mentioned above.

Emerging alternative approaches to countering antibiotic resistance have revealed the antibacterial power of essential oils^{9,10}.

The use of essential oils is receiving increasing attention due to their intrinsic antimicrobial properties. The hypothesis of a synergy between antibiotics and essential oils opens promising prospects in the fight against bacterial resistance¹¹.

In this study, we chose *Escherichia coli* as our bacterial model because of its clinical relevance and its ability to represent other pathogenic enterobacteria sharing similar resistance mechanisms. We evaluate the antibacterial activity of essential oils, alone and in combination with antibiotics, to explore their synergistic potential.

EOs have gained increasing attention for their antimicrobial properties and potential to enhance the efficacy of conventional antibiotics. Derived from aromatic plants, EOs contain a complex mixture of bioactive compounds that can disrupt bacterial membranes, inhibit biofilm formation, and modulate microbial resistance mechanisms^{12,13}. In this study, we selected Tea Tree (*Melaleuca alternifolia*), Clove (*Syzygium aromaticum*), Eucalyptus (*Eucalyptus globulus*),

Thyme (*Thymus vulgaris*), and Oregano (*Origanum vulgare*) essential oils based on their well-documented antibacterial activities^{12,14–19}. These essential oils have been extensively studied for their ability to target a broad spectrum of pathogens, making them promising candidates for combination therapies with antibiotics.

The results obtained are discussed in terms of possible new therapeutic strategies to combat resistant bacteria.

MATERIAL AND METHOD

CULTURE MEDIA

Mueller Hinton agar (BIOKAR, ref. BK048HA) was prepared at a concentration of 38 g/L in Milli-Q water. The solution was autoclaved at 120°C for 20 minutes and poured into sterile Petri dishes under aseptic conditions. Each dish contained approximately 25 mL of solidified media. Luria Bertani (LB) was also prepared by dissolving 20 g/L in Milli-Q water and autoclaving using the same protocol.

BACTERIAL CULTURE

A single colony of *E. Coli* was taken and inoculated into 5 mL of LB and was then incubated at 37°C. The resulting bacterial suspension was adjusted to an optical density of 0.5 at 600 nm, which corresponds to approximately $5 \cdot 10^8$ CFU/mL. This suspension was diluted 0,5 times to get a final concentration of 2.5×10^8 CFU/mL for the antimicrobial activity test.

ANTIMICROBIAL ACTIVITY TEST

The essential oils of Tea Tree, Clove, Thymol, Oregano, Eucalyptus have antibacterial and/or antiseptic properties. These pure essential oils tested were diluted in paraffine oil. Final concentrations were prepared by serial dilution in a 96-

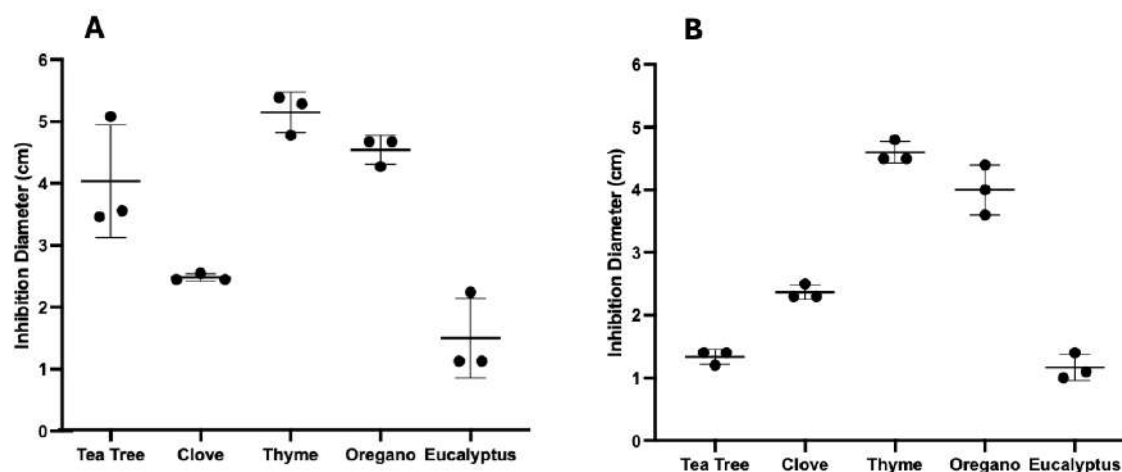


Figure 2: Determination of diameter of inhibition zones for pure (A) and 1/2 dilution (B) of essential oils against *E. coli*. The figure illustrates the inhibition zones of *E. coli* treated with different essential oils, both in their pure form (A) and at 1/2 dilution (B). The inhibition zones were measured in centimeters and are expressed as mean values with error bars representing standard deviations. Statistical analysis using the Kruskal-Wallis test indicated a significant difference between groups in both conditions ($p < 0.001$ for pure essential oils, $p < 0.0001$ for diluted oils). However, Mann-Whitney pairwise comparisons revealed no significant differences between specific pairs of pure essential oils ($p > 0.05$). The results suggest that different essential oils exhibit varying antibacterial activities against *E. coli*, with the inhibition capacity being influenced by concentration.

well plate. For each oil, 20 μ L were placed on a disc. Then, the discs were put on the agar plates previously inoculated with *E. coli* suspension. 5 discs per plate were placed and all the tests were carried out in triplicate (Figure 1). The plates were incubated at 37°C for 24 hours. The minimum inhibitory concentration was determined by measuring the diameter of inhibition zones around the discs and comparing the results with controls.

CONTROLS

Discs were impregnated with 20 μ L of each control. Then, the discs were put on agar plates previously inoculated with *E. coli* suspension. The positive control is composed of 20 μ L of diluted ampicillin at 10 μ g/mL, concentration at which the bacteria are effectively eradicated, impregnated on a disc. The negative control is composed of 20 μ L of sterile water impregnated on a disc. The control's plates were incubated in the same condition as plates for the antimicrobial activity test.

SYNERGY DETERMINATION

Two essential oils (Thymol, Oregano) were selected for this study. The essential oils were diluted in paraffin oil. Serial dilutions were carried out in a 96-well plate to achieve the desired final concentrations. For comparison, essential oils were diluted to 50% of their initial concentration (1/2 dilution).

Antibiotics were tested at two concentrations: 5% and 10%. These concentrations were prepared by diluting the antibiotic in water to achieve the desired final concentrations for the experiments.

For each essential oil and antibiotic, 20 μ L of the antibiotic and 20 μ L of mixed EO was placed on a sterile paper disc. These discs were then placed on agar²⁰ plates that had been previously inoculated with an *E. coli* suspension. All tests were performed in triplicate. The plates were incubated at 37°C for 24 hours.

The diameter of the inhibition zones around the discs was measured to determine the minimum inhibitory concentration (MIC) for each treatment. Synergy was determined by comparing the effects of essential oils alone and in combination with antibiotics at 5% and 10%, using the green (largest inhibition zone) and red (smallest inhibition zone) lines on the graph as reference points.

The thresholds were defined based on the average inhibition zones of the individual essential oils. Specifically, we calculated two reference values: a lower bound (mean low) and an upper bound (mean high), corresponding to the average inhibition zones of each essential oil when tested alone.

A combination was considered synergistic if its inhibition zone exceeded the upper threshold, antagonistic if it fell below the lower threshold, and neutral (no synergy) if it remained between the two thresholds. This approach allows for a standardized comparison of the combined effects relative to the individual essential oils.

STATISTICAL TESTS

To determine if there is statistical significance between the measured inhibition zone diameters, various statistical tests

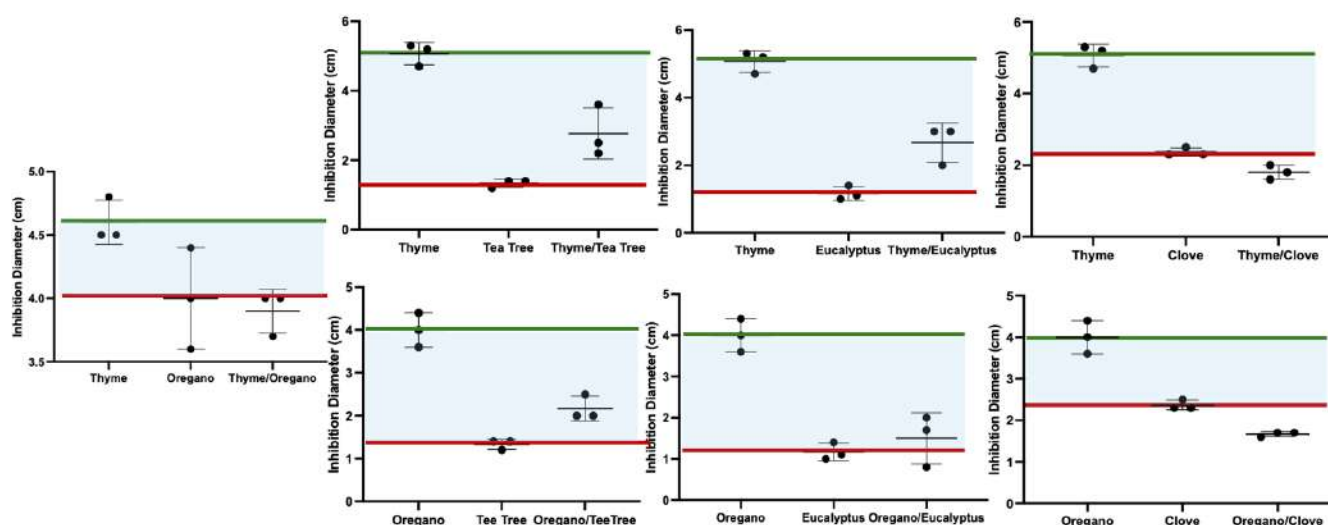


Figure 3: Evaluation of synergies between essential oils. The figure illustrates the evaluation of synergies between different essential oil combinations against *E. coli*, based on inhibition zone measurements. The synergy assessment was conducted by comparing the combined inhibition zones to two reference thresholds: a lower bound (mean low) and an upper bound (mean high), calculated from the mean inhibition zones of individual essential oils. The tested combinations included Thyme/Oregano, Thyme/Tea Tree, Oregano/Tea Tree, Thyme/Eucalyptus, Oregano/Eucalyptus, Thyme/Clove, and Oregano/Clove. Synergistic effects (green) were observed when the inhibition zone exceeded the upper bound, indicating an enhanced antibacterial activity. No synergy (blue) was reported when the inhibition zone remained between the two thresholds, suggesting an additive or independent effect. Antagonistic interactions (red) were identified when the inhibition zone fell below the lower bound, indicating a reduction in antibacterial activity. Each combination was tested in triplicate, and results are presented as mean inhibition zones with error bars representing standard deviations.

were used. The Kruskal-Wallis test was applied to compare the diameters of inhibition zones among the different essential oils. The Mann-Whitney test was also employed to compare the effectiveness of each essential oil with the others. The choice of these tests is justified as the Kruskal-Wallis test is suitable for comparing multiple groups with non-normally distributed data, while the Mann-Whitney test is ideal for comparing two independent groups to assess differences in effectiveness between essential oil pairs. Additionally, a post-hoc test was conducted to identify which specific pairs of groups exhibited significant differences. The significance threshold for all tests was set at 0.05% ($p < 0.05$). All statistical analyses were performed using GraphPad software.

RESULTS

ANTIBACTERIAL ACTIVITY OF SELECTED ESSENTIAL OILS AGAINST *E. COLI*

First, we determine the inhibitory capacities of different essential oils, both in their pure form and at a 1:2 dilution. We observed that each essential oil had a distinct level of efficacy.

Figure 2a compares the inhibition diameters of different pure essential oils on *E. coli*. Thyme (mean = 5.1 cm) and oregano (mean = 4.5 cm) showed the largest zones of inhibition. Two statistical tests were then performed: the Kruskal-Wallis test revealed a significant difference among the five essential oils, but the Mann-Whitney test showed no significant differences when comparing the essential oils pairwise.

Figure 2b illustrates the inhibition diameters of the same essential oils, this time diluted at a 1:2 ratio. The results were like those in Figure 2a, with thyme and oregano again showing the largest inhibition diameters (4.5 cm and 4.0 cm, respectively). The same statistical tests yielded the same outcomes as in Figure 2a.

Overall, these results indicate that thyme and oregano are the most effective essential oils against *E. coli* under our experimental conditions. Despite the lack of significance in the pairwise comparisons (Mann-Whitney test), the focus of subsequent investigations will be on these two essential oils.

ANTIBACTERIAL ACTIVITY OF SELECTED ESSENTIAL OILS AGAINST *E. COLI*

Figure 3 shows that the combinations Thyme/Clove, Oregano/Clove, and Thyme/Oregano have inhibition zones below the lower threshold, exhibiting an antagonist effect. However, all the other combinations (Thyme/Tea Tree, Thyme/Eucalyptus, Oregano/Tea Tree, and Oregano/Eucalyptus) have inhibition zones between both the upper and lower thresholds. This indicates an absence of a synergistic effect. We have not observed any synergy between the essential oils tested. The results of this analysis are summarized in Table 2.

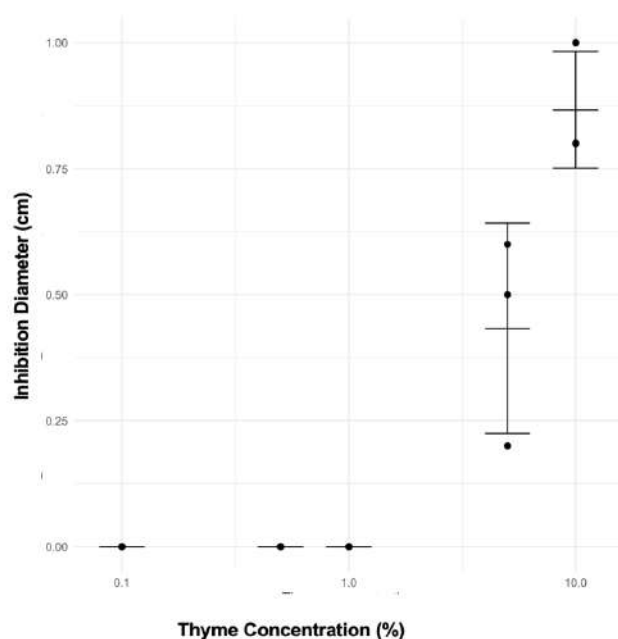


Figure 4: Minimum inhibitory concentration (MIC) determination of thyme essential oil against *Escherichia coli* using agar diffusion method. The figure illustrates the minimum inhibitory concentration (MIC) of thyme essential oil tested against *E. coli* on solid medium (agar diffusion method). The MIC was determined to be 5%, indicating the lowest concentration of thyme essential oil that completely inhibits the growth of *E. coli*. Each concentration was tested in triplicate, and the results are expressed as mean inhibition zones with error bars representing standard deviations. The MIC value (5%) corresponds to the concentration at which no visible bacterial growth was observed around the disks after 24 hours of incubation at 37°C.

DETERMINATION OF MIC EOS

Figure 4 presented with a logarithmic scale of concentrations, shows a dose-dependent relationship between the concentration of thyme essential oil and its activity against *Escherichia coli*. For concentrations below 0.1%, no inhibition is observed. Slight activity begins to appear at 0.1%, but significant inhibition is only observed at 5%, with a diameter of approximately 0.7 cm, which further increases at 10% (1.2 cm).

We identify a minimum inhibitory concentration (MIC) at 5% (5µL for 100µL), corresponding to the lowest concentration that completely prevents bacterial growth. The error bars indicate good reproducibility of the measurements performed in triplicate.

These results confirm that thyme essential oil exhibits effective antibacterial activity starting at 50µL/mL highlighting its potential as a natural antimicrobial agent.

SYNERGISTIC ACTIVITY

In **Figure 5**, we wanted to show if a synergy between the more efficient essential oil tested (thyme) and an antibiotic (Ampicillin here) was possible. We tested with the best amount of thyme oil (5%) and with two different percentages of Ampicillin, 5 and 10 %.

For the panel A, the mean of the inhibition diameter of the 5% thyme is around 0,42 cm and the one for the 10% Ampicillin is around 0,17cm. Then, when we put the two together the mean is around 0,19cm, so we see that it is comprised between the mean values of thyme and ampicillin alone ($0,17 < 0,19 < 0,42$).

For the panel B, the mean of the inhibition diameter of the 5% thyme is still around 0,42cm and the one for the 5% Ampicillin is around 0,14 cm.

We see that the values of “Thyme 5% / Amp 10%” and “Thyme 5% / Amp 5%” are within the range defined above. This means that there is no synergy between the 5% and 10% antibiotics and thyme essential oil. Furthermore, after carrying out the Kruskal-Wallis statistical test, we found no significant difference between the different groups tested. There was no change in the inhibitory effect, so no synergy.

Table 1 presents the composition of each essential oil used for our study. From the supplier site²¹.

DISCUSSION

The aim of this study was to highlight and confirm the bactericidal or bacteriostatic action of relevant essential oils selected from the literature.

The results of minimum inhibitory concentration measurements for each essential oil against the *E. coli* model confirmed the antibacterial potential of several essential oils, particularly thyme and oregano (**Figure 2**). These oils were therefore selected for synergy tests with the antibiotic. These two oils showed the highest inhibition diameters, although statistical tests were not always able to demonstrate a significant difference, which corroborates the results of previous articles that have already demonstrated the efficacy of these oils^{13,22}. The insignificance statistically could be explained by important disparity between triplicate results of a same oil. We suppose that this could be due to the ability of the oil to diffuse in the medium (not uniform), creating differences in inhibition zones. Or this might come from experimental errors, such as inconsistent application of the oil or differences in drop size which influenced the results.

According to the components of essential oils listed in Table 1, the presence of terpenes in thyme and oregano oils, which are responsible for the formation of phenols such as thymol (present in thyme and oregano) and carvacrol (present in oregano), could be one of the reasons for the effectiveness of

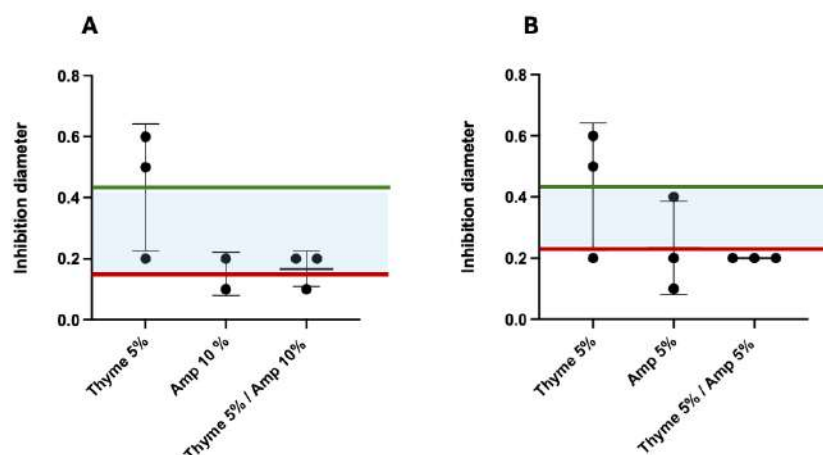


Figure 5: Inhibition zone comparison to evaluate the synergy between thyme and ampicillin 10% (A) and 5% (B) The figure presents a comparative analysis of inhibition zones to assess the potential synergy between thyme essential oil and ampicillin at two concentrations (10% in panel A and 5% in panel B). A Kruskal-Wallis test was performed to determine statistical differences among the tested groups. The analysis yielded a Kruskal-Wallis statistic of 3.74 with a p-value of 0.24 for panel A and a Kruskal-Wallis statistic of 2.88 with a p-value of 0.43 for panel B, indicating no statistically significant differences between the inhibition zones. These results suggest that the combination of thyme essential oil with ampicillin at 10% and 5% does not lead to a significant enhancement or reduction in antibacterial activity. This implies the absence of a synergistic or antagonistic interaction at these concentrations. Each condition was tested in triplicate, and the results are represented as mean inhibition zones with error bars indicating standard deviations.

the bactericidal effect of these oils. These two compounds act as membrane disruptors²³. However, we must also notice that the essential oils used in our experiment are coming from different origin and could also have different composition and extraction methods (Table 1). It would have been interesting to test essential oils of thyme and oregano coming from another supplier. Due to financial constraints, we were unable to do that. However, we suppose that the difference of composition of a same essential oil might have an impact on the experiment.

In our tests with combined essential oils, no synergy was found either with the antibiotic or between two essential oils. However, we did observe an antagonistic effect when the oils were combined with clove (Figure 3). This antagonism could be attributed to potential interactions between the compounds in the essential oils and clove, which may have reduced their effectiveness. Further studies with different concentrations and additional combinations of essential oils are needed to better understand these interactions.

The minimum inhibitory concentration (MIC) of thyme was determined at 5%, i.e., 50 $\mu\text{L/mL}$. We observed a MIC for thyme fairly similar to other studies, ranging from 6 $\mu\text{L/mL}$ to 50 $\mu\text{L/mL}$ ²⁴ (Figure 4). It should be kept in mind that the MIC varies depending on the supplier of the essential oil and on the bacterial strain chosen. Determining the MIC is essential for considering potential future therapeutic applications. However, it will probably have to be reconsidered in the light of the limits associated with the use of essential oils in high concentrations (risk of irritation, toxicity, other undesirable effects).

It is important to note that we encountered difficulties when using our positive controls containing ampicillin. We did not observe any zone of inhibition around the tablet for this normally positive control, whereas we observed zones of inhibition for tablets with essential oils. However, this did not pose a problem, as the zones of inhibition around the tablets for the tests with essential oils enabled the manipulation to be validated; the situation would have been different if all the tests had been negative: the experiment would then have had to be repeated. The antibiotic prepared for the positive control may have been too diluted during preparation, or the disc may have been poorly soaked.

There was also an inconsistency, with the less concentrated Ampicillin (5%) being more effective than the more concentrated Ampicillin (10%) (Figure 4). This could be due to an error in the preparation of the antibiotic (Figure 5).

The results of the study highlight the complexity of the interactions between the active molecules in essential oils and the need to better understand the mechanisms behind these effects. More in-depth research is needed to gain a better understanding of how the oils interact with the bacteria and to pave the way for new strategies to combat antibiotic resistance. We could also have carried out synergy tests with oils that showed less bactericidal activity. Other essential oils could also be tested to determine synergistic effects with oils that have already proved effective against bacteria. In-depth determination of the molecular composition of the oils could also be envisaged to understand in greater detail the action of the many active ingredients presents in essential oils. Studies on other bacterial models would improve the relevance of the

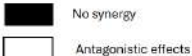
trials, given the varied capacities of bacteria to resist the action of antibiotics.

Table 1: References and composition of the essential oils used for the experiments.

	origin	active principle
Tea tree	Part of the plant: Leaves Method of extraction: steam distillation Origin: South Africa and Zimbabwe Culture: Biologic	Terpinène < 48%, Gamma-terpinène < 28%, 1.8 cinéole < 15%, Alpha terpinène < 13%, Paracymène < 12%, Alpha terpineole < 8%, Alpha pinène < 6%, Terpinène 4-ol < 5%, D-Limonène < 4%, Béta pinène < 3%, Myrcène < 3%.
Clove	Parti of the plant: Clove Method of extraction: steam distillation Origin : Madagascar culture : biologic	Eugénol < 87% Eugenyl acetate < 15% Beta carophyllène < 7%
Oregano	Part of the plant : aerial part Method of extraction: steam distillation Origin : Spain Culture : Biologic	Carvacrol Thymol
Thyme	Part of the plant : flowering heads Method of extraction : steam distillation Origin: Central Asia Culture : Conventional	Terpinene-4-ol Trans-Thuyanol Linalol Limonène
Eucalyptus	Part of the plant : Leaves Method of extraction : steam distillation Origin : Centrale Asia Culture : Conventional	Eucalyptol limonène

Table 2: Synergy evaluation matrix for essential oils. The matrix highlights the interactions between the essential oils, showing antagonistic effects and no synergy, but no evidence of positive synergy

	Thyme	Oregano	Tea Tree	Clove	Eucalyptus
Thyme					
Oregano					



CONCLUSION

In conclusion of our study, we can observe no synergy effect between any essential oil or with antibiotic. This has been verified graphically and statistically with several statistical tests by a p-value under 0,05 each time. Nevertheless, our study has enabled us to demonstrate the antimicrobial effect of thyme and oregano and to highlight the Minimum Inhibited Concentration (MIC) of thyme which is around 5% of essential oil.

So, we can see that essential oils are an interesting alternative to antibiotic treatment and that is a promising solution in the fight against antibiotic resistance.

However, further research is needed to determine the best essential oil to use against resistant bacteria. In addition, the effectiveness of these essential oils needs to be tested on human cells and then in vivo. It would be possible to carry out initial efficacy tests on mouse models¹¹.

In conclusion, our study opens new prospects in the fight against antibiotic resistance, a global health problem, using essential oils, particularly those of thyme and oregano.

CONFLICTS OF INTEREST

The authors declare no potential conflicts of interest related to this study.

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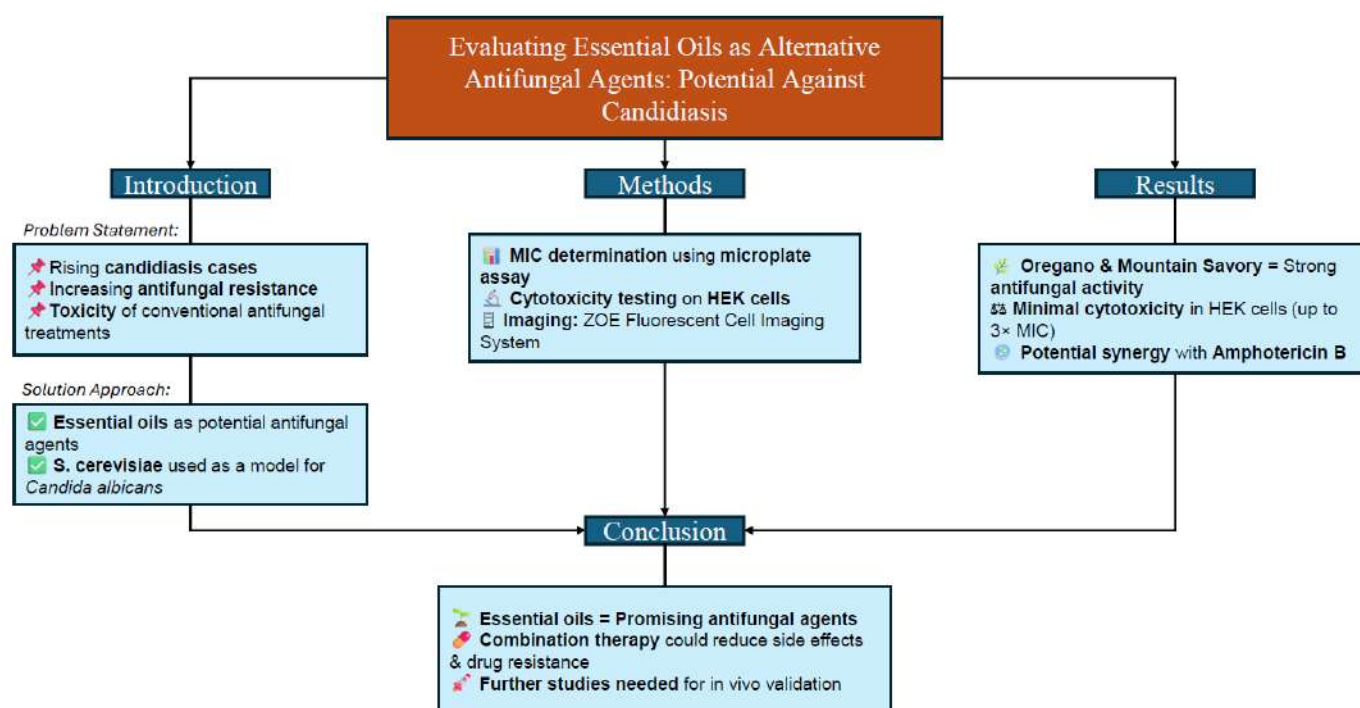
Evaluating essential oils as alternative antifungal agents: potential against candidiasis

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VISUAL ABSTRACT



Evaluating essential oils as alternative antifungal agents: potential against candidiasis

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Abstract – The increasing prevalence of candidiasis, coupled with the growing concern over antifungal resistance and the severe side effects of current treatments, underscores the urgent need to explore alternative therapies. This study assesses the antifungal potential of essential oils, including spearmint, grapefruit, basil, lavender, tea tree, mountain savory, oregano, and red thyme, against *Saccharomyces cerevisiae*, used as a model for *Candida albicans*, the leading cause of candidiasis.

The minimum inhibitory concentration (MIC) was determined through a microplate assay. Results showed that oregano and mountain savory essential oils exhibited strong antifungal activity, significantly inhibiting *S. cerevisiae* growth. Cytotoxicity assays indicated no significant cell death in human HEK cells treated with concentrations of the oils up to three times the MIC, suggesting a favorable safety profile for eukaryotic cells. Nevertheless, further studies are necessary to quantify and confirm the non-cytotoxicity of these essential oils in HEK cells.

The findings suggest that oregano and savory essential oils hold promise as alternative antifungal agents with minimal toxicity. Their integration into combination therapies could potentially enhance the efficacy of candidiasis treatment while reducing reliance on high doses of conventional antifungal drugs, thus mitigating side effects, and addressing resistance challenges.

Index Terms – *Saccharomyces cerevisiae*, Candidiasis, Antifungal resistance, Essential oils, Amphotericin B, minimum inhibitory concentration.

INTRODUCTION

Candidiasis, a fungal infection caused predominantly by *Candida* species, represents a significant health concern,

especially in immunocompromised individuals¹. This opportunistic pathogen can lead to infections ranging from superficial, such as oral and vaginal thrush, to more severe forms like invasive candidiasis², affecting various organs including the pulmonary epithelium. According to recent epidemiological data³, candidiasis accounts for a substantial burden on healthcare systems, with rising incidences worldwide^{1,2,4}. The challenges associated with treating candidiasis stem from both the pathogen's ability to form biofilms and its increasing resistance to conventional antifungal agents⁵.

The increase in cancer, diabetes, and HIV is leading to a rise in respiratory fungal diseases⁴. Furthermore, according to the World Health Organization, 300 million people worldwide suffer from asthma⁶, which is largely due to allergic reactions to airborne fungi. Current treatment regimens, while effective in some cases, come with considerable drawbacks. Many antifungal therapies, such as azoles (Fluconazole, Itraconazole, Voriconazole), polyenes (Amphotericin B) and echinocandin (Anidulafungin, Caspofungin, Micafungin) exhibit serious side effects⁷⁻⁹, including hepatotoxicity and nephrotoxicity, and are often limited by reduced efficacy against resistant strains¹⁰. These limitations underscore the urgent need for novel therapeutic alternatives that are both effective and safer for patients.

In recent years, essential oils have garnered growing interest as potential antifungal agents¹¹. These natural compounds are known for their diverse bioactive properties^{12,13}, low cost, minimal side effects, and ease of production and administration. Preliminary studies have already indicated promising antifungal activities of certain essential oils¹⁴⁻¹⁸ against *Candida* species, raising the possibility of their use as a viable treatment option for candidiasis.

For the purposes of this study, we utilized *Saccharomyces cerevisiae* as a model organism¹⁹ instead of *Candida* species.

S. cerevisiae, though non-pathogenic, shares key cellular and molecular characteristics with *Candida*, making it a widely accepted surrogate for studying antifungal agents. Its well-characterized genetics and ease of manipulation provide significant experimental advantages, allowing for the rapid screening of antifungal effects.

In parallel, Human Embryonic Kidney (HEK) 293 cells are commonly used as a model of human cells²⁰. Indeed, HEK cells are frequently used in biological and pharmacological research due to their reliable growth characteristics, human origin, and susceptibility to various compounds^{20,21}. Their use in this study facilitates the evaluation of the potential cytotoxicity of essential oils on human cells, thereby offering a comprehensive approach to assessing both efficacy and safety.

Given the potential advantages of essential oils, and the context of our lab, this study aimed to identify and prove essential oils effectiveness in the context of pulmonary candidiasis, by using *S. cerevisiae* and HEK cells as models. The goal was to confirm findings from previous studies^{14–18}, extend understanding of essential oil mechanisms, and contribute to the development of safer and more effective treatment options for fungal diseases.

MATERIAL AND METHOD

MATERIALS

Chemicals

We used essential oils as mentioned in the **Error! Reference source not found.** below.

Table 1: Summary of essential oils used.

Essential oils	Marketplace	Serial Number
Spearmint	VOSHUILES.com	B06XZCRV78
Grapefruit	Puressentiel	CIP: 6010475
Basil	Puressentiel	CIP: 9942270
Lavender	Puressentiel	CIP: 9942525
Tea Tree	Puressentiel	CIP: 9942258
Mountain Savory	Puressentiel	CIP: 6010469
Oregano	Puressentiel	CIP: 9945400
Red Thyme	VOSHUILES.com	B01KTQ8DDK

We worked with *Saccharomyces cerevisiae* and HEK293 cells from SupBiotech. For culture, we used Sabouraud agar medium (BIOKAR BK025HA) and Mueller-Hinton Broth (BIOKAR BK108HA) supplemented with 2% (v/v) of Tween20 (SupBiotech) for yeast culture and MEM (SIGMA D6546-500ML) supplemented with 10% (v/v) FBS (SIGMA F7524-100ML) and 1% (v/v) penicillin + streptomycin (SIGMA P4333-100ML) for HEK293 cells. And, Trypan Blue solution (SIGMA T8154-20ML), DPBS (SIGMA D8537-500ML) and Trypsin/EDTA (SIGMA T4174-100ML).

We used 2% (v/v) of Tween 20 to dissolve essential oils as they have low solubility in water.

Organism and Cell Culture

We precultured *S. cerevisiae* in Sabouraud agar medium at 28-30°C in a static incubator for 24 hours, then transferred at 0,5 Mc Farland into 25mL of Mueller-Hinton Broth.

We cultured HEK293 cells with MEM supplemented with 10% (v/v) FBS and 1% (v/v) penicillin + streptomycin at 37°C in a humidified atmosphere with 5% CO₂ in a static incubator. Cells were seeded in a T25 flask at 4x10⁴ cell/cm² in 9mL final.

ANTIFUNGAL ACTIVITY ASSAY

We evaluated the antifungal activity of essential oils against *S. cerevisiae* using a broth microdilution method.

For each essential oil, we used a P100 micropipette with a sterile tip to introduce 100 µL of Mueller-Hinton broth (with 2% (v/v) Tween 20) into wells 1 to 12 of a 96-well plate. Using the same tip, we added 100 µL of the pure essential oil solution into well 1 and mixed the contents thoroughly by aspirating and dispensing three times. After mixing, we discarded the tip and used a new sterile tip to transfer 100 µL from well 1 to well 2, performing the first dilution. We repeated this serial dilution process from well 2 to well 12, discarding 100 µL from well 12 into a bleach solution after the final dilution.

This serial dilution was performed at a 1:4 ratio, resulting in final concentrations of 25%, 12.5%, 6.25%, 3.13%, 1.56%, 0.78%, 0.39%, 0.2%, 0.1%, 0.05%, 0.02%, and 0.01% in duplicate as shown in **Error! Reference source not found.**

Table 2: minimum inhibitory concentration (MIC) protocol summary.

Oil	1	2	3	4	5	6	7	8	9	10	11	12
MH Broth	100	100	100	100	100	100	100	100	100	100	100	100
Essential Oil	100	100	100	100	100	100	100	100	100	100	100	100
<i>S. cerevisiae</i>	100	100	100	100	100	100	100	100	100	100	100	100

We included controls in the experiment as follows. First, a negative control A consisting of 100 µL of culture medium, which serves as a baseline to confirm the absence of any external contamination or background interference. Secondly, a negative control B consisting of 100 µL of the essential oil mixed with 100 µL of culture medium for each oil, to test the sterility and inherent properties of the oils without the presence of yeast. Then a positive control consisting of 100 µL of the inoculum in 100 µL of culture medium, to verify the expected growth behavior under standard conditions. We repeated this protocol for each essential oil, following the plate layout shown in **Error! Reference source not found.**

Following the preparation of the controls and test samples, we incubated each plate at 30°C for 24 hours, providing optimal

Table 3: Minimum inhibitory concentration (MIC) plate layout.

96 wells plate	1	2	3	4	5	6	7	8	9	10	11	12
A	25,00%	12,50%	6,25%	3,13%	1,56%	0,78%	0,39%	0,20%	0,10%	0,05%	0,02%	0,01%
B												
C	T+	T-										
D												
E	25,00%	12,50%	6,25%	3,13%	1,56%	0,78%	0,39%	0,20%	0,10%	0,05%	0,02%	0,01%
F												
G	T+	T-										
H												

conditions for yeast growth. After incubation, we measured growth in each well using a microplate reader, the ZOE Fluorescent Cell Imaging System from Bio-Rad. We determined the minimum inhibitory concentration (MIC) as the lowest concentration of essential oil that inhibited 90% of yeast growth compared to the positive control, as shown in **Error! Reference source not found..**

ensure their viability. For the samples, we prepared one flask for each essential oil and treatment duration (24 hours/48 hours). We selected four essential oils based on their MIC results.

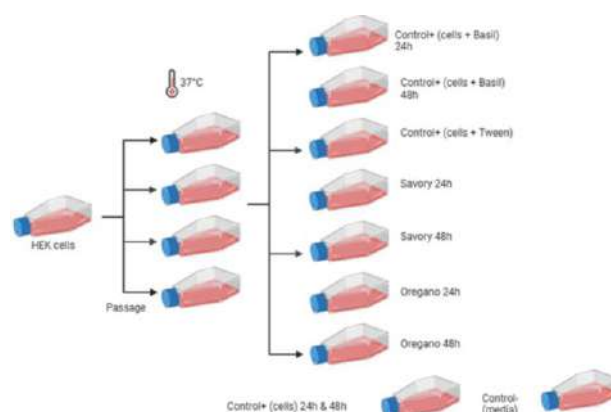


Figure 2: Cytotoxicity assessment protocol for essential oils of oregano and savory.

Figure 1: MIC protocol: determination of minimum inhibitory concentration.

This critical value provides a quantitative measure of the antifungal efficacy of each essential oil, serving as a benchmark for evaluating their potential use as antifungal agents. These controls and methodologies collectively ensured the reliability and accuracy of the experimental outcomes.

CYTOTOXICITY ASSAY

We initiated precultures from our frozen stock of HEK cells. Using the master stock, we prepared two T25 flasks and incubated the cells at 37°C with 5–10% CO₂. Once the cultures reached 80% confluence, we counted the cells and passaged them by seeding T25 flasks at a density of 4×10⁴ cells/cm², which resulted in 1×10⁶ cells per flask (=25×4×10⁴). Based on the cell count, we added the appropriate volume of cells to each flask, adjusting to a total volume of 9 mL of complete MEM.

For seeding, we placed the cells in 9 mL of complete medium in T25 culture flasks. We prepared 10 flasks, as shown in **Error! Reference source not found..** We prepared 2 control flasks (for 24-hour and 48-hour time points), one with only the media to ensure its sterility; the second with only cells to

We began treatment 24 hours after seeding by adding 1 mL of culture medium containing the essential oil dissolved in 2% (v/v) Tween 20 at the desired concentration to each flask. We adjusted the final concentration to 3 times the MIC, accounting for two dilution steps: dilution of the essential oil in 1 mL of medium, followed by dilution of this 1 mL into the 9 mL of culture medium.

After 24 or 48 hours of treatment, we assessed cell confluence. We rinsed the cells twice with 5 mL of DPBS, discarding the supernatant each time. To detach the cells, we added 1 mL of 0.01% Trypsin/EDTA without discarding the supernatant. We neutralized the trypsin action by adding 5 mL of culture medium. Using an inverted microscope, we confirmed cell detachment and facilitated it by gently tapping the flask. We determined cell viability and the total cell count using trypan blue staining. We prepared a 1:1 dilution of 50 µL of the cell suspension with 50 µL of trypan blue (adjusting the volume based on cell concentration) and counted the cells using a hemocytometer.

RESULTS

I. HEDONIC TEST METHOD

This study focused on a hedonic test designed to assess sensory

ANTIFUNGAL ACTIVITY RESULTS

In the **Error! Reference source not found.**, the absence of cell growth in the negative controls and the presence of robust growth in the positive control validate the reliability of the results, confirming that the experimental setup was free from contamination and that the yeast maintained its viability under standard conditions. These controls serve as critical benchmarks, ensuring that any observed inhibitory effects on cell growth can be attributed solely to the test conditions, or the compounds being evaluated. This validation strengthens the credibility of the findings and provides a solid foundation for interpreting the effectiveness of the essential oils as antifungal agents.

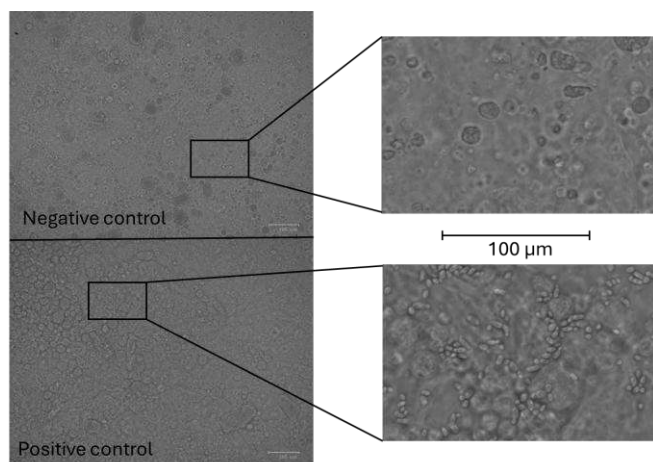


Figure 3: Zoe microscope photograph of “positive control” and “negative control” wells (zoom: 175x). In the negative control (MH broth only), we can observe an absence of microorganisms. The round shapes we see are caused by the Tween reagent. In the positive control (MH broth + *S. cerevisiae*), we can observe the presence of *S. cerevisiae*.

Microscopic observation using the ZOE Fluorescent Cell Imaging System enables precise assessment of cell growth by comparing the visual characteristics of the test samples to the established patterns of negative and positive controls. In the **Error! Reference source not found.**, the case of savory essential oil, the analysis revealed an absence of cell growth at ~0.39% of essential oil, characterized by a lack of confluency. In contrast, the well at ~0.20% of essential oil displayed clear evidence of cell growth, with robust cell confluency comparable to that of the positive control. A similar trend was observed for oregano essential oil, where at ~0.39% showed no detectable cell growth, while we observed a visually strong cellular proliferation at ~0.20% of essential oil. These findings suggest a concentration-dependent effect of the

essential oils, providing critical insights into their cytotoxic profiles.

Among the tested essential oils, basil, grapefruit, and spearmint demonstrated no effect on the cell growth, indicating they are not effective against *S. cerevisiae* as they do not have antifungal activities. In contrast, tea tree and lavender exhibited moderate activity with MIC values of 25% and 12.50%, respectively. Red thyme showed higher MICs at 6.25%. However, savory and oregano demonstrated the lowest MICs (0.39%), indicating the highest antifungal activity against *S. cerevisiae*. The results in **Error! Reference source not found.** indicate that savory and oregano are the most potent antifungal agents under the tested conditions.

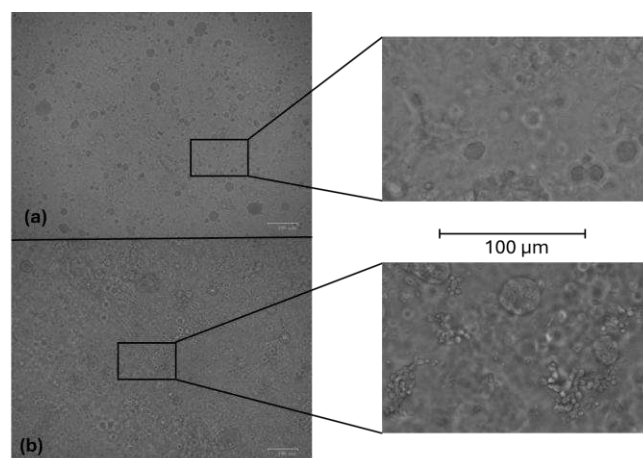


Figure 4: Zoe microscope photograph of wells 7 (no growth) and 8 (growth) for the savory essential oil sample (zoom: 175x). (a) At ~0.39% of savory essential oil, we can observe an absence of microorganisms. (b) At ~0.20% of savory essential oil, we can observe the presence of *S. cerevisiae*. (The round shapes we see are caused by the Tween reagent).

Given their potent antifungal effects, savory and oregano essential oils could be considered promising candidates for the development of natural antifungal agents, particularly in situations where conventional antifungal treatments may be ineffective or toxic. In this idea, we evaluated the cytotoxicity of these oils to determine their potential as therapeutic agents.

CYTOTOXICITY RESULTS

Based on the MIC results, we conducted the cytotoxicity study using oregano and savory essential oils. We adjusted the final essential oil concentration to three times the MIC, as shown in **Error! Reference source not found.**

First, the positive controls were effective, as HEK cells demonstrated robust growth under the tested conditions with no discernible morphological changes. This confirms that the cells were viable, and the culture environment was conducive to their proliferation. The negative controls showed no cell growth, verifying that there was no contamination.

To evaluate the cytotoxicity of our samples, we will refer to ISO 10993-5²², which states that “only a reduction in the measurement value greater than 30% is considered a cytotoxic effect. A reduction of less than 30% is therefore considered to fall within the uncertainty of the method.”

Table 3: Minimum inhibitory concentration (MIC) results. The grey squares with the minus symbol represent the inhibited wells with no cell growth. The essential oils written in bold are the most effective and the most interesting to study. The white squares with plus symbol represent the uninhibited wells where *S. cerevisiae* have grown. (DF: Dilution factor)

Oil	1	2	3	4	5	6	7	8	9	10	11	12
Basil	+	+	+	+	+	+	+	+	+	+	+	+
Red thyme	-	-	-	+	+	+	+	+	+	+	+	+
Grapefruit	+	+	+	+	+	+	+	+	+	+	+	+
Teatree	-	+	+	+	+	+	+	+	+	+	+	+
Savory	-	-	-	-	-	-	-	+	+	+	+	+
Origano	-	-	-	-	-	-	-	+	+	+	+	+
Lavender	-	-	+	+	+	+	+	+	+	+	+	+
Peppermint	+	+	+	+	+	+	+	+	+	+	+	+
DF	4	8	16	32	64	128	256	512	1024	2048	4096	8192
[C] % oil	25,00	12,50	6,25	3,13	1,56	0,78	0,39	0,20	0,10	0,05	0,02	0,01

Microscopic observation with the ZOE system allowed us to assess cell confluence and determine only visually whether the essential oils exhibited cytotoxic effects or not.

Table 4: Cytotoxicity assay protocol.

	MIC results		Cytotoxicity				
	MIC (%oil)	3xMIC (%oil)	Oil Volume (μL)	Tween20 (2%) Volume (μL)	MEM volume (μL)	%oil in 1mL	%oil in 10mL final
Savory	0,39	1,172	105	20	875	11,73	1,173
Origano	0,39	1,172	105	20	875	11,73	1,173
Control- (Tween)			0	20	980	0	0

In the **Error! Reference source not found.**, we observed bubble formation in the presence of Tween 20, indicating that Tween 20 at 2% (v/v) caused bubbles, which slightly hindered visual observation under the ZOE Fluorescent Cell Imaging System. Moreover, a morphological difference was noted in cells exposed to Tween 20, as they appeared round and non-adherent. This observation was also made for cells treated with three times the MIC of both origano and savory essential oils, likely due to the presence of Tween 20 in those samples as well.

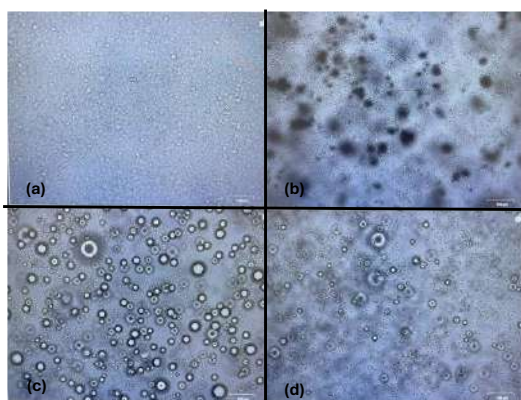


Figure 5: Zoe microscope photograph of cytotoxicity results for origano and savory essential oils (zoom: 175x). (a) HEK cells with only MEM. We can observe a cell mat with high confluency. (b) HEK cells with MEM and Tween 20. We can observe deformed and detached cells. (c) and (d) HEK cells with MEM, Tween 20 and savory (c) or origano (d) essential oil. We can observe deformed and

detached cells with approximately the same confluency as (b).

However, an error in cell counting using trypan blue staining prevents the quantification of the non-cytotoxicity of the two essential oils. Indeed, we observed no cells dead or alive in the Malassez counting chamber.

We can visually distinguish healthy cells in the positive control (HEK cells with MEM) by their polygonal shape and adherence to the surface. In contrast, HEK cells exposed to Tween 20 appear round and non-adherent. However, a similar cell shape and density are observed between HEK cells treated with Tween 20 alone and those treated with Tween 20 in the presence of three times the MIC of origano and savory essential oils. This difference is solely due to the presence of Tween 20, which was used instead of Sorbitan due to a shipment issue.

DISCUSSION

The alarming rise in antifungal resistance and the increasing prevalence of fungal infections, particularly candidiasis caused by *Candida albicans*, demand innovative therapeutic approaches. This study highlights the potential of essential oils as alternative antifungal agents, using *Saccharomyces cerevisiae* model organism to simulate *Candida albicans* due to its similar growth characteristics and ease of handling.

First, we aimed to evaluate the antifungal activity of essential oils against *S. cerevisiae*. To achieve this, we tested various essential oils using the broth microdilution method. Among the tested essential oils, origano (*Origanum vulgare*) and mountain savory (*Satureja montana*) demonstrated remarkable antifungal activity, as evidenced by their low minimum inhibitor concentrations (MICs). Our results are consistent with previous studies^{11,16} on the antifungal properties of essential oils such as origano and red thyme. The potent antifungal activity observed with savory and origano, however, suggests that these oils may hold

significant potential for further exploration. The antifungal activity of essential oils is likely due to their ability to disrupt cell membranes or interfere with fungal metabolism, though further research is needed to elucidate these mechanisms in *S. cerevisiae*. However, some differences are noticeable between previous studies^{11,16} and our own. Basil showed no effect on *S. cerevisiae* whereas it showed antimicrobial effect in the other studies that used it against *Candida albicans*. This could be explained by certain differences in their cell wall structure between these two species^{23–25}. Further tests should therefore be undertaken using oregano and savory against *C. albicans*.

The use of essential oils as antifungal agents is promising not only because of their broad-spectrum activity but also because they offer a more natural alternative to synthetic drugs, which have heavy side effects. However, their hydrophobic nature and potential cytotoxicity remain challenges.

Since no differences were observed between cells treated with Tween 20 alone and those treated with Tween 20 and essential oils, we can hypothesize that the essential oils do not exhibit cytotoxic effects on HEK cells. However, due to issues with cell counting, these results cannot be conclusively interpreted. Our cytotoxicity assays may have revealed no significant impact on human HEK cells at concentrations three times the MIC, suggesting that these oils may be safe at therapeutic doses. Nevertheless, even with previous studies^{14–18,26–28} that have demonstrated the non-cytotoxicity of the tested essential oils, supporting their potential safety for use, an additional study should be conducted to determine, quantify, and verify the potential non-cytotoxicity of these essential oils, at the specified concentrations, on HEK cells. Sorbitan should be used as an alternative to Tween 20 to avoid any potential morphological changes in HEK cells.

Combination therapies have been a cornerstone of resistance management in antifungal treatments. Amphotericin B, a widely used antifungal, is associated with severe side effects, including nephrotoxicity. In this study, in the annex, we also explored the potential synergy between essential oils and amphotericin B. Preliminary results suggest that these combinations could allow for reduced doses of amphotericin B, potentially mitigating its adverse effects while maintaining therapeutic efficacy. However, more comprehensive studies are needed to validate these findings and determine the exact mechanisms of interaction between the essential oils and amphotericin B.

The study faced some limitations, such as the need to confirm the non-cytotoxicity of essential oils with more quantitative data and the need to evaluate directly on *Candida albicans*. Additionally, the role of solvents such as Tween 20 in influencing results warrants further investigation. The use of sorbitan-based emulsifiers is a more suitable alternative to avoid morphological changes in cells.

CONCLUSION

This project demonstrates the promising antifungal potential of oregano and mountain savory essential oils against *Saccharomyces cerevisiae* as a model for *Candida albicans*. Their notable antifungal activity and potential non-cytotoxicity effects on human cells position them as viable candidates for future antifungal treatments. The observed synergy with amphotericin B underscores the possibility of reducing the side effects associated with traditional antifungal therapies, thereby addressing one of the major challenges in candidiasis management.

Future work should focus on expanding these findings through in vivo studies, mechanistic evaluations of essential oil interactions with antifungal drugs, and assessments of their efficacy against clinical strains of *Candida albicans*. This multidisciplinary approach could pave the way for the development of safer, more effective, and resistance-combating antifungal therapies.

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Highlighting the therapeutic potential of thymol essential oil against Aspergillosis, a fungal respiratory disease

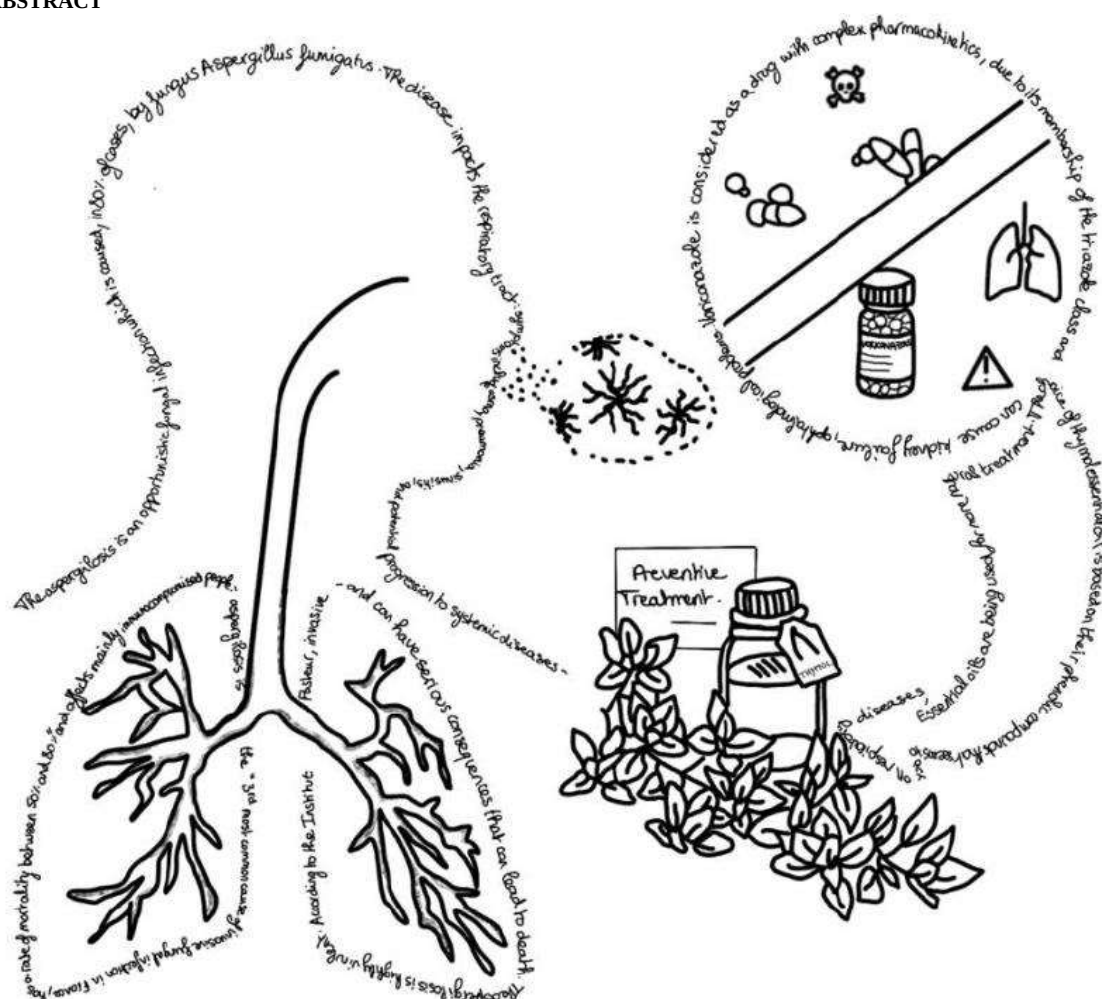
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VISUAL ABSTRACT



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Highlighting the therapeutic potential of thymol essential oil against Aspergillosis, a fungal respiratory disease

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Abstract – The aim of this study is to find a remedy based on essential oil for fungal respiratory diseases. The aspergillosis is an opportunistic fungal infection caused most often by *Aspergillus fumigatus*. Here we are focusing on respiratory damages caused by this fungus. Based on literature we have selected the thymol essential oil as the most effective treatment, due to its high concentration in phenolic compounds. The purpose of our article is to demonstrate that thymol essential oil is a promising candidate for the preventive treatment of aspergillosis. We have used *Aspergillus Oryzae* to modelized *Aspergillus Fumigatus* and HEK 293 cells to modelized pulmonary cells, because of laboratory and material availability constraints. We have first determined the MIC (Minimal Inhibition Concentration) of thymol essential oils on *Aspergillus Oryzae*. Then we have tested the cytotoxicity of thymol essential oil on HEK 293 cells, which have no impact on cellular viability. Finally, we have tested the effectiveness of Thymol essential oil in condition of hybrid culture. We have cultivated HEK 293 cells in a medium enriched with thymol essential oil and have added *Aspergillus Oryzae* when cells were confluent. No cytotoxicity was observed for cells and a mycelium apparition was observed in the medium after 48h of incubation with Thymol essential oil. From those results we can conclude that the thymol essential oil did not prevent fungus growth but only slowed it at concentration tested.

Index Terms – Aspergillosis, Cells, Fungus, Minimal Inhibition Concentration, Thymol essential oil, Treatment.

INTRODUCTION

The aspergillosis is an opportunistic fungal infection which is caused, in 80% of cases, by the fungus *Aspergillus fumigatus*. Its life cycle can be viewed in the figure 1 “Natural cycle of

Aspergillus in the environment”. The cycle is continuous: The first phase consists of the emission of airborne spores, followed by deposition and germination on an organic substrate. The mycelium filaments begin to develop, forming a vegetative mycelium and finally an asexual reproductive system. Aspergillosis caused by this fungus has usually an impact on the respiratory tract: symptoms include asthma, pneumonia, sinusitis and potential progression to systemic diseases [1].

The aspergillosis is highly virulent. According to the Institut Pasteur, invasive aspergillosis is the “3rd most common cause of invasive fungal infection in France, has a mortality rate of between 50% and 80%” and affects mainly immunocompromised individuals [1]. Today, it is one of the most feared infectious diseases, due to the emergence of strains that are increasingly resistant to treatment, and the toxicity of the treatments used. [1]

Although this disease affects a large number of people and can have serious consequences that can lead to death, no truly effective treatment, without side effects, has yet been discovered. Currently, different drugs are used such as Isavuconazole, Posaconazole, Itraconazole or Voriconazole, which is one of the most effective treatments against Aspergillosis, with “53% response to treatment”. [2] [3] However, Voriconazole is considered as a drug with complex pharmacokinetics, due to its membership of the triazole class [3]. Triazoles are compounds with a high risk of toxicity (cardiac, neurological and visual disorders), as well as variability of action from one individual to another. What's more, they are highly sensitive, interacting with other drugs such as anti-allergy treatments, tuberculosis treatments or epileptic seizures. [4] A drug of this type will act on the fungus cell membrane, and more specifically on ergosterol, to induce cell death by dysfunction. Voriconazole has also a

number of contraindications and can cause kidney failure, ophthalmological problems, psychiatric problems and, in some cases, even cytotoxicity, thereby increasing the likelihood of death [5].

This lack of effective treatment leads us to turn to essential oils, which are increasingly being used for more natural treatments. The objective is to find a treatment based on essential oils, which can act and be effective, without the undesirable effects of current treatments. Our 2023 Market study “*Mise en évidence du potentiel thérapeutique d'huiles essentielles contre des maladies respiratoires fongiques*” (“*Highlighting the therapeutic potential of essential oils against fungal respiratory diseases*”) revealed that a preventive medical treatment would be the most interesting form to develop. The criteria of accessibility and potential were evaluated and considered within the study to determine whether the most interesting drug to produce came from the medical or paramedical field and from a preventive or curative use.

The choice of thymol essential oil is based on the finding of Mónica Zuzarte et Lúgia Salgueiroce:

“Overall, essential oils rich in phenolic compounds seem to be very effective but clinical translation requires more comprehensive *in vivo* studies and human trials to assess the efficacy and tolerability of these compounds in respiratory mycosis.”, “Once again phenolic compounds stand out as very effective, namely eugenol—MIC = 250 µg/mL, thymol—MIC = 192 µg/mL and carvacrol—MIC = 0.16 µL/mL.” [6]

Phenolic compounds are composed of an aromatic hydrocarbon ring with one or more hydroxide groups attached and seem to act on respiratory diseases: The more phenolic compounds an essential oil contains, the more effective it is. The presence or absence of phenolic compounds is determined by the MIC (Minimum Inhibitory Concentration), which represents the minimum concentration of the drug needed to inhibit fungal growth [6]. The MIC index, expressed in µg/mL, corresponds in our case, to the lowest concentration of essential oil at which the growth of our strain is no longer visible.

Thymol oil, with an MIC of 192 µg/mL according to the literature, once again appears to be the ideal candidate [6]. This article therefore focuses on thymol essential oil and more specifically on the following question:

Is thymol essential oil a promising candidate as a preventive treatment against aspergillosis?

To answer this question, we chose to conduct our experiments on *Aspergillus oryzae*. This decision was motivated by the fact that we did not have a P2 biosafety level laboratory. As *Aspergillus oryzae* is a lower biosafety level organism, this allowed us to conduct our research safely and in accordance

with our available resources. Our experiments were also carried out on a HEK 293 cell model. The cell type was chosen for availability and ease of handling.

MATERIAL AND METHOD

DETERMINATION OF THE MIC OF THE THYMOL ESSENTIAL OIL ON *ASPERGILLUS ORYZAE*

With an indicative MIC of 192 µg/mL, and an initial oil concentration of 44.8% for 200 µL, thymol essential oil was diluted by 50th with unenriched Eagle's medium [6].

Diluted thymol essential oil was used to do a cascade dilution range in 12 test tubes (SUPBIOTECH), with 2 mL of the previous tube being introduced each time into 2 mL of unenriched Eagle media and 1 mL of *Aspergillus Oryzae* suspension with a concentration of 106 spores/mL. Thymol essential oil concentrations were ranged from 17203.2 µg/mL to 0 µg/mL, and triplicates (3 ranges) were done for each concentration.

Then, 1 mL of each range was transferred to a 24-well plate, and two ranges were transferred to Falcons tubes under agitation (for material reasons), all three being kept at 28°C for 24 hours. For all manipulations and volume transfers, sterile equipment, Bunsen beakers, and graduated pipettes were used.

After this experiment on *Aspergillus Oryzae* we want to ensure that the MIC of the Thymol essential oil have no impact on the viability of HEK 293 cells.

VIABILITY TEST ON HEK 293 CELLS

The first step was the defrosting of the cells. The cells vial was soaked in a double boiler at 37°C for 30 sec to 1 minute. Then, the content was put drop by drop into 9 mL of medium Eagle with (FCS (10%) + L-Glu (0.5%)). The cells suspension was centrifuged for 5 minutes at 0.2 g at room temperature between 22 and 25°C. 5 mL of Eagle's medium (enriched) was added to re-suspend the cells by gentle pipetting. A little part (15 µL) was used to count the cell viability at the beginning. This stage was done under a functional PSM (type 2 microbiological safety cabinet) previously disinfected with Surfanios.

The cells were counted with Malassez cell.

To use it, the cells were diluted (with an adapted dilution) and coloured. To do this, blue trypan was used, which stains dead cells blue and leaves living cells uncoloured.

The Malassez cell is composed of one glass slide with two hollows. To observe cells, a strip was put on the top of the glass slide, and the liquid was deposited between the strip and the glass slide. In the middle of the glass slide, there were two hollows that formed a grid, which was used to count the cells. This grid was composed of 20 squares, and each square had a

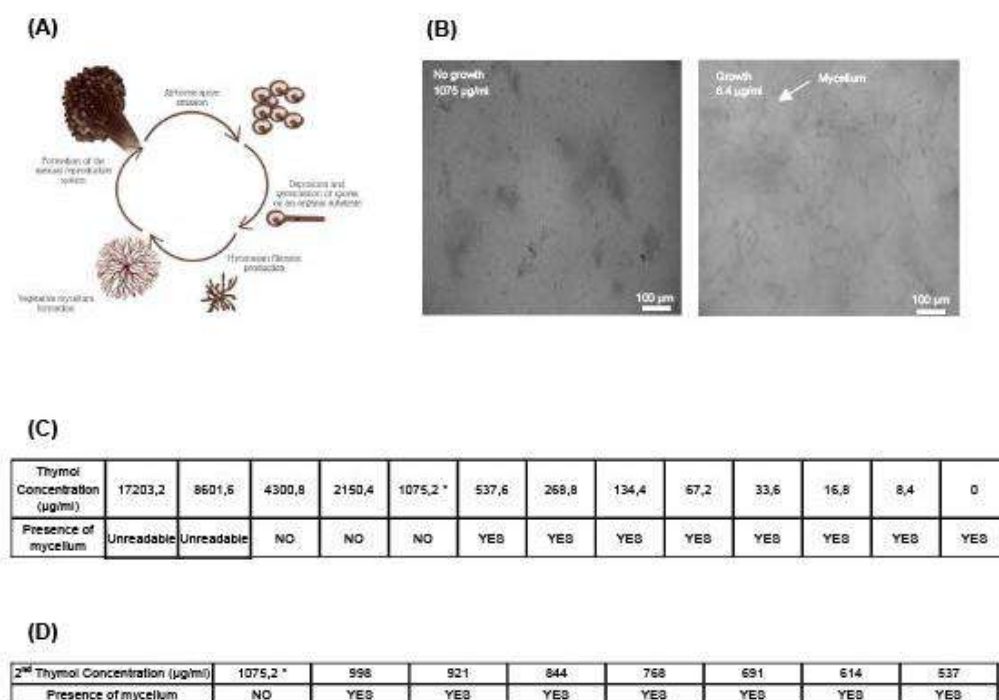


Figure 1: Determining MIC of the Thymol Essential Oil looking at *Aspergillus Oryzae* development. (A) Natural cycle of *Aspergillus* in the environment adapted of Gangneux J.-P, Bouchara J.-P, Chabasse D. *Encycl Méd Chir* (Elsevier, Paris). *Maladies infectieuses*, 8-600-A-10, 2013. (B) Fungus do not grow in the presence of thymol essential oil (1075µg/ml) (photo left) and mushrooms grow in the presence of thymol essential oil (8.4µg/ml) (photo right). (C) MIC Thymol table (* MIC concentration). (D) MIC Thymol table more precise (* MIC concentration).

surface area of 0.01 mm³, corresponding to 0.01 µL. To calculate the number of cells, a number of squares was determined to count each cell, and the number of alive cells and dead cells was differentiated. Then, the following formula was used [7]:

$$N = \frac{n}{a.v.Fd}$$

With:

- N the number of cells per unit volume
- n the number of counted cells
- a the number of units counted
- v the volume of a metering unit
- Fd the dilution factor

For the cells growth, the cells suspension was put in a flask of 25 cm² with 5 mL of enriched Eagle's medium (corresponding to Eagle's medium to which 10% FBS (Foetal Bovine Serum) was added to provide growth factors and proteins required for cell growth, 0.5% L-Glutamine for proper cell function, notably for protein and glucose creation, and 5% antibiotic to avoid contamination). Then, the cells were put at 37°C in an incubator with 5% of CO₂.

Once in culture, the cells were monitored and observed under a light microscope (10x or 20x objective) to determine cell

mat confluence for adherent cells, the presence of debris and/or suspended cells.

After confluence was reached, cell passage was required. The elements used for cell passage were Enriched Eagle's medium, PBS (phosphate-buffered saline) as a cleaning solution, and Trypsin to detach adherent cells from flanges.

This stage was done under a functional PSM (type 2 microbiological safety cabinet) previously disinfected with Surfanios, as all materials used under the PSM were.

Enriched Eagle's medium, PBS, and trypsin-0.5% EDTA were placed in a 37°C water bath at least 15 minutes before use. 5 mL of PBS was used as a wash buffer and was dispensed into the flasks to remove all traces of FBS.

The wash medium was removed by aspiration, and 5 mL of trypsin-EDTA was introduced to loosen the cell mat, then the flask was incubated for a few minutes at 37°C. The inactivation of the trypsin action was achieved by further introduction of FBS in the form of 5 mL of complete medium.

The tube was centrifuged for 5 minutes at 0.2 g at room temperature between 22°C and 25°C. Then, the supernatant from the flask was removed by aspiration or inverting, depending on how the pellet behaved. The cell pellet was then

re-suspended by gentle pipetting with 5 mL of complete medium.

To test the sensibility of the cells with essential oil, a lot of cells could be used for the experiment after cell passages to multiply cells. A 6 well-plate was seeded with cells (77,545 cells/well). 4 days maximum were waited, and the medium was changed. One day after the medium change, Thymol essential oil was added at 11 different concentrations in triplicate [Figure 1c].

Five other wells were added for the control. The negative control with just cells (2 wells) and the positive one with a drop of pure essential oil (3 wells).

24 hours were waited, and the cell viability was calculated with the Malassez cell.

To get closer to reality in a human body and to see the comportment of *Aspergillus Oryzae*, HEK 293 and Thymol essential oil together, an hybrid culture was took in place.

HYBRID CULTURE

For the mixed cell culture of HEK 293 cells and *Aspergillus Oryzae* fungi, it was necessary to work under the PSM when manipulating the cells before the final observation. 8 24-well plates were prepared in total, to observe cells and fungus at different times: 0 (« T0 »), 15 (« T15 »), 24 (« T24 »), and 48 hours (« T48 »), with 2 plates for each period. Using 2 plates avoided mixing conditions « with » and « without » essential oils to prevent any contamination (the different conditions are detailed in the result part).

For each well, the HEK 293 cells were seeded with 500 μ L of enriched Eagle medium, containing or not thymol essential oil depending on the condition of each well at T0. Then, T24 and T48 were incubated at 37°C for 3 hours. After the incubation, the fungus was incorporated, the medium was removed, and 250 μ L of medium with fungus ($c=106$ cells/ml) and 250 μ L of medium ($C_{\text{medium}} \times 2$ for conditions with essential oils to keep the concentrations as initial) were added. For the condition without fungus, only 250 μ L of normal medium was added. For T0, T15, and T24, the plates were observed under an optical microscope to see if there was a development of fungus spores. For T48, in addition to optical observation, the cells viability ratio was counted using Malassez cells [8].

With all those experiments, conclusion can be drawn in the result part.

RESULTS

Following the material and methods steps, here are presented the results.

DETERMINATION OF THE MIC OF THE THYMOL OIL ON A.ORYZAE

The presence or absence of mycelium were observed in wells for thymol oil concentrations ranging from 0 μ g/ml to 17203.2 μ g/mL [Figure 1b].

By tacking the concentration in an increasing way of essential oil, the presence of mycelium was observed from 0 μ g/mL to 537.6 μ g/mL.

From 1075.2 μ g/mL there is no longer any presence of mycelium. The results were not readable for the two highest concentrations (17203.2 μ g/mL and 8601.6 μ g/mL) linked to the thickness of the oil preventing a clear observation in the well.

By observing these results and thus for which concentrations the mycelium is present or absent, the MIC value was deduced which is 1075.2 μ g/mL.

VIABILITY TEST ON HEK 293 CELLS

The essential oil sensibility was tested on cells. As a reminder, the cells were incubated in contact with the essential oil to measure their viability.

The viability observed were greater than 85% in all wells, regardless of the essential oil concentration.

Therefore, it was possible to deduce that for concentrations of thymol essential oils less than or equal to 17203.2 μ g/mL, thymol essential oil has no cytotoxic effect on HEK 293 cells.

HYBRID CULTURE

After determining the MIC for fungi and observing the effect of essential oil on mycelial development and cell viability, the fungi and cells were combined in the same culture.

Multiple conditions were tested.

For the condition 1 we put cells + medium

For the condition 2 we put cells + medium + essential oil

For the condition 3 (positive control) we put cells + fungi

For the condition 4 we put cells + medium + essential oil MIC + fungi

For the condition 4bis we did the same as condition 4 with MIC $\times 2$

For the condition 5 we put fungi + medium + essential oil

And for the condition 6 we put fungi + medium.

At T=0: we seeded the plates and incubated.

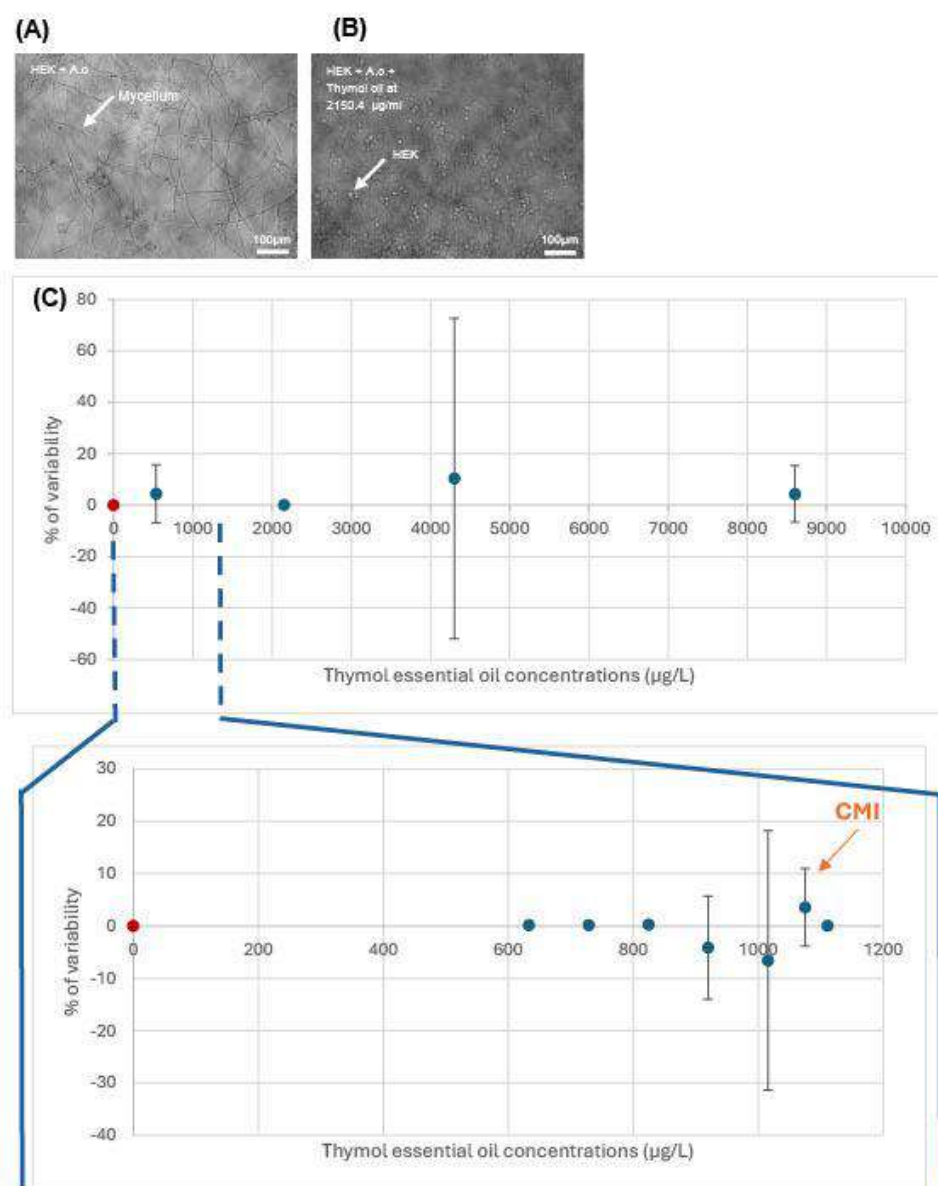


Figure 2: Sandwich culture. (A) Positive controls (HEK293 + *Aspergillus Oryzae* cells, $t=24h$, without essential oil). (B) Mixed culture (HEK293 + *Aspergillus Oryzae*, $t=24h$, thymol oil at MIC x2). (C) Viability difference of HEK cells in the presence of thymol essential oil versus negative control (red dot). Calculated for points: x-axis: Concentration in thymol essential oil; y-axis: Represent the gap between the cell viability in the test condition relative to the negative control with the formula: $[\text{mean variability of negative control}] - [\text{mean variability of the point}]$. When the standard deviation is not readable it is because there are insignificant.

At $T=24$: the presence of mycelium was observed in all the wells where fungi were seeded except those at MICx2 [figures 2a, 2b, 2c].

At $T=48$: It was possible to see the appearance of mycelium in the wells at MICx2.

So the essential oil slowed down the fungus growth but didn't stop it. Some conditions in experiments can explain that.

DISCUSSION

These initial results are certainly relatively promising in terms of efficiency and capacity to reduce cytotoxicity. This potential essential oil-based treatment has enabled us to understand the relationship and the impact of thymol essential oil on the fungal and cellular environment. However, we should keep in mind that inhibition is not complete in our experiments, given that the growth of the fungus in hybrid culture is slowed but not completely stopped. Furthermore,

having observed that the hybrid culture has no cytotoxic effect up to 48 h after incubation, we have no visibility considering the evolution of the different compounds after this period.

Moreover, the difference between our obtained MIC and the literature MIC is due to several reasons: the experiments made in the literature were run with pure extracted phenolic compounds from thymol essential oils, whereas our experiments were with complete essential oils. The culture medium that we used for fungus to determine the thymol essential oil was an enriched medium, because it was the same as the one for cells culture. This culture medium may have encouraged the growth of fungi, making it more resistant to thymol essential oil. The MIC in literature was determined with *Aspergillus Fumigatus*, while in our experiments the MIC was determined with *Aspergillus Oryzae*. The MIC difference could be explained by the strain difference between our study and the literature.

Following this three-year project, it might be relevant to test these same experiments on a model close to the real conditions of aspergillosis. Indeed, tests will be essential to ensure that the results are also conclusive on a lung cell model as well as the original strain *Aspergillus Fumigatus*, and then, ultimately, in an in vivo model.

In addition, it is important to note that some standard deviations may be due to the fact that the experiments were sometimes carried out by different members of the group, which may lead to differences in handling. It should also be noted that the essential oil of Thymol, presenting a thick oily layer on certain results, and despite our vigilance, sometimes caused variation in the precision of the microscopic observations.

Furthermore, as the disease mainly affects immunocompromised people, it would be interesting to consider a curative treatment to stop the disease after the fungus has developed in the system.

CONCLUSION

Aspergillosis, a fungal respiratory disease, affects tens of thousands of people worldwide every year. However, current treatment does not guarantee a cure, and in half of cases results in death due to cytotoxicity.

In this study, the aim was to propose an alternative treatment that would be effective but non-toxic, providing a less risky solution for patients.

Following a literature review, we selected thymol essential oil for its antifungal and supposedly non-cytotoxic properties against cells. As far as the strain of fungus was concerned, to get as close as possible to the strain responsible for the disease, we chose to study the *Aspergillus Oryzae* strain, which, unlike the original *Aspergillus Fumigatus* strain, could be handled in the laboratory.

Following a marketing study and considering the regulations and the needs of patients, we opted for a preventive treatment.

The results of our experiments led to the following conclusions:

Under thymol essential oil and *Aspergillus Oryzae* conditions only, with a MIC of 1075 µg/mL, thymol essential oil inhibits the growth of the strain at this concentration and above.

In HEK 293 cells, cell viability was maintained at a thymol essential oil concentration of less than 17.203 µg/mL.

In the mixed in vitro culture (HEK 293 and *Aspergillus Oryzae* in the presence of thymol essential oil), our results suggested that thymol did not prevent fungal growth but only slowed the strain's growth rate. However, cell viability was not affected.

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